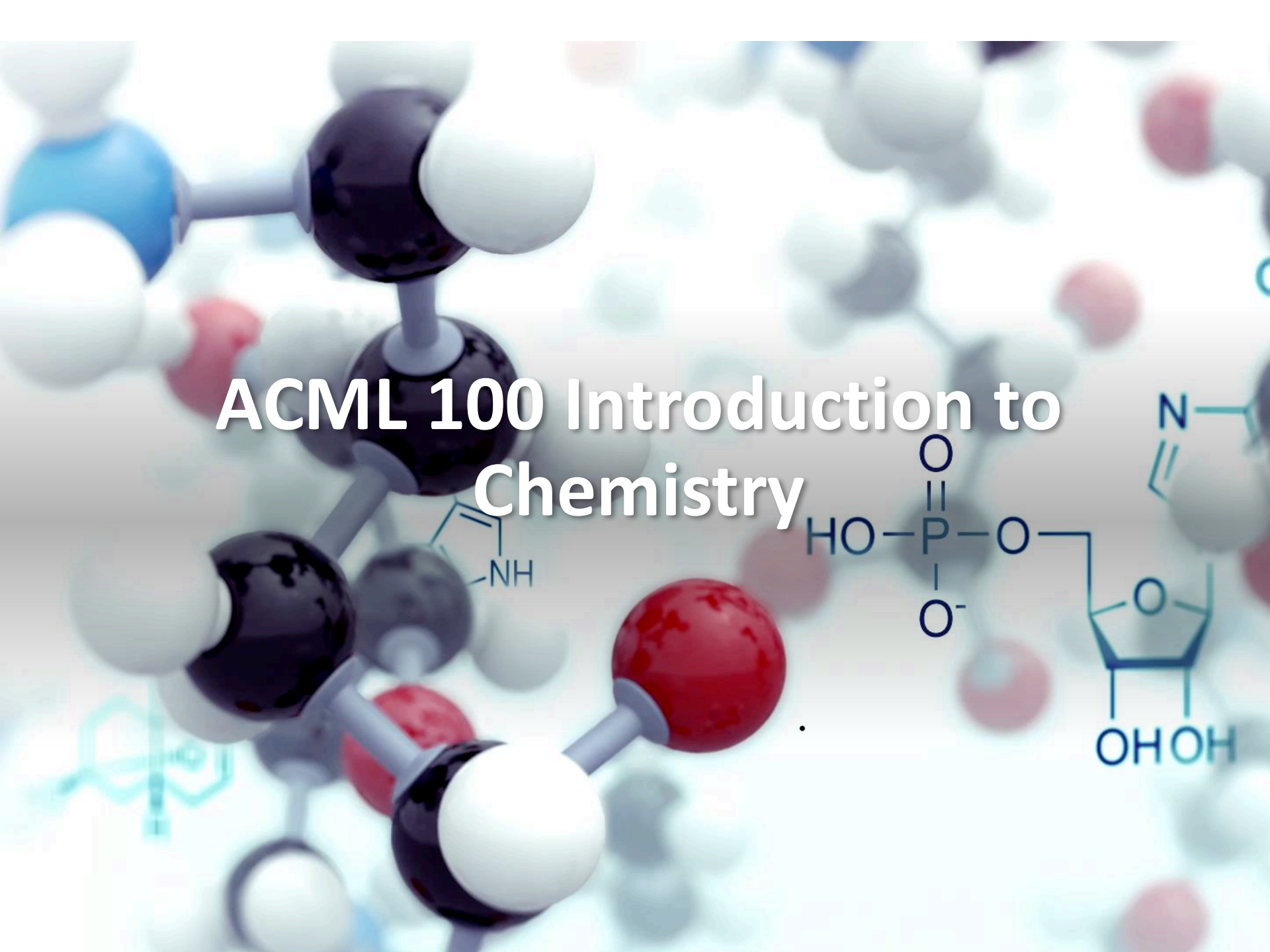
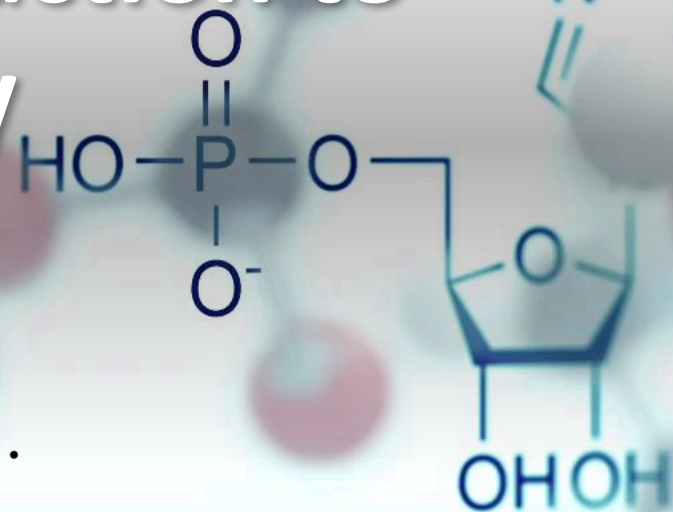
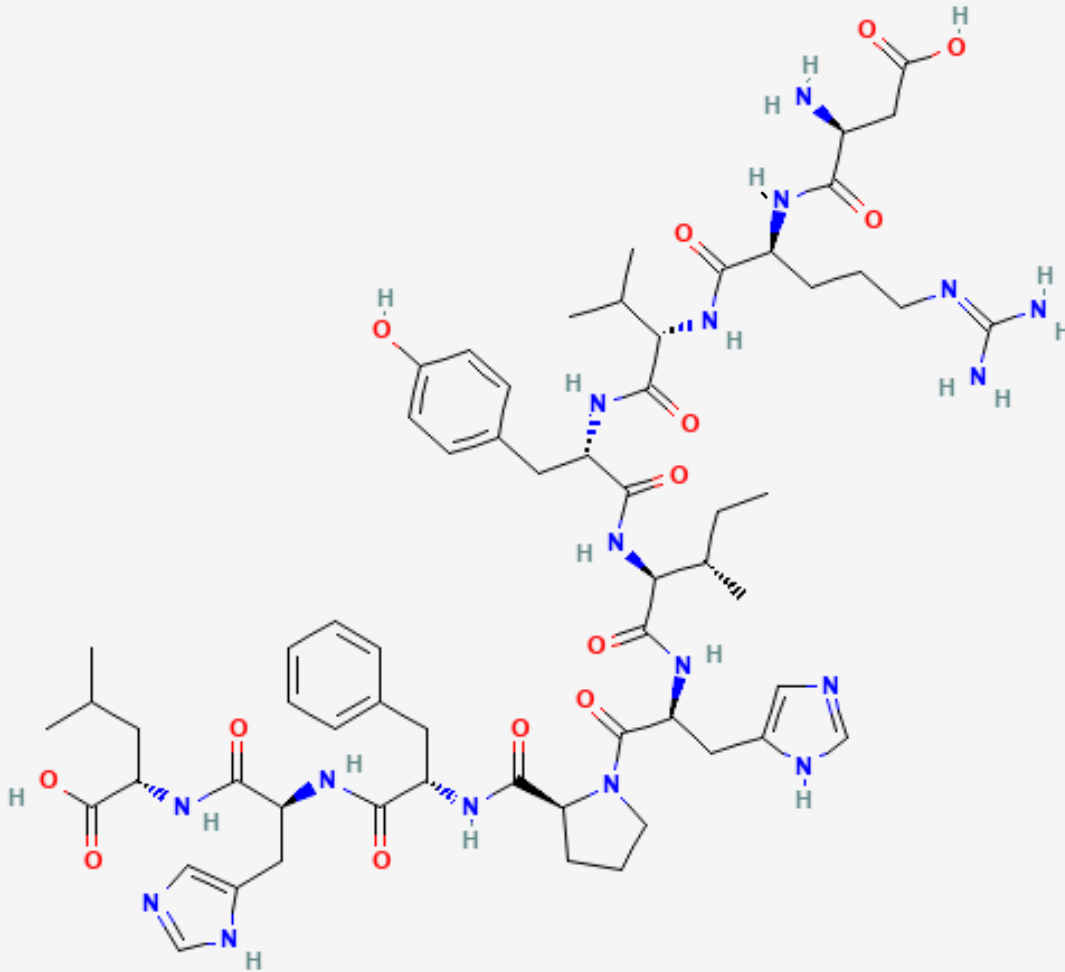




# ACML 100 Introduction to Chemistry



# Angiotensin



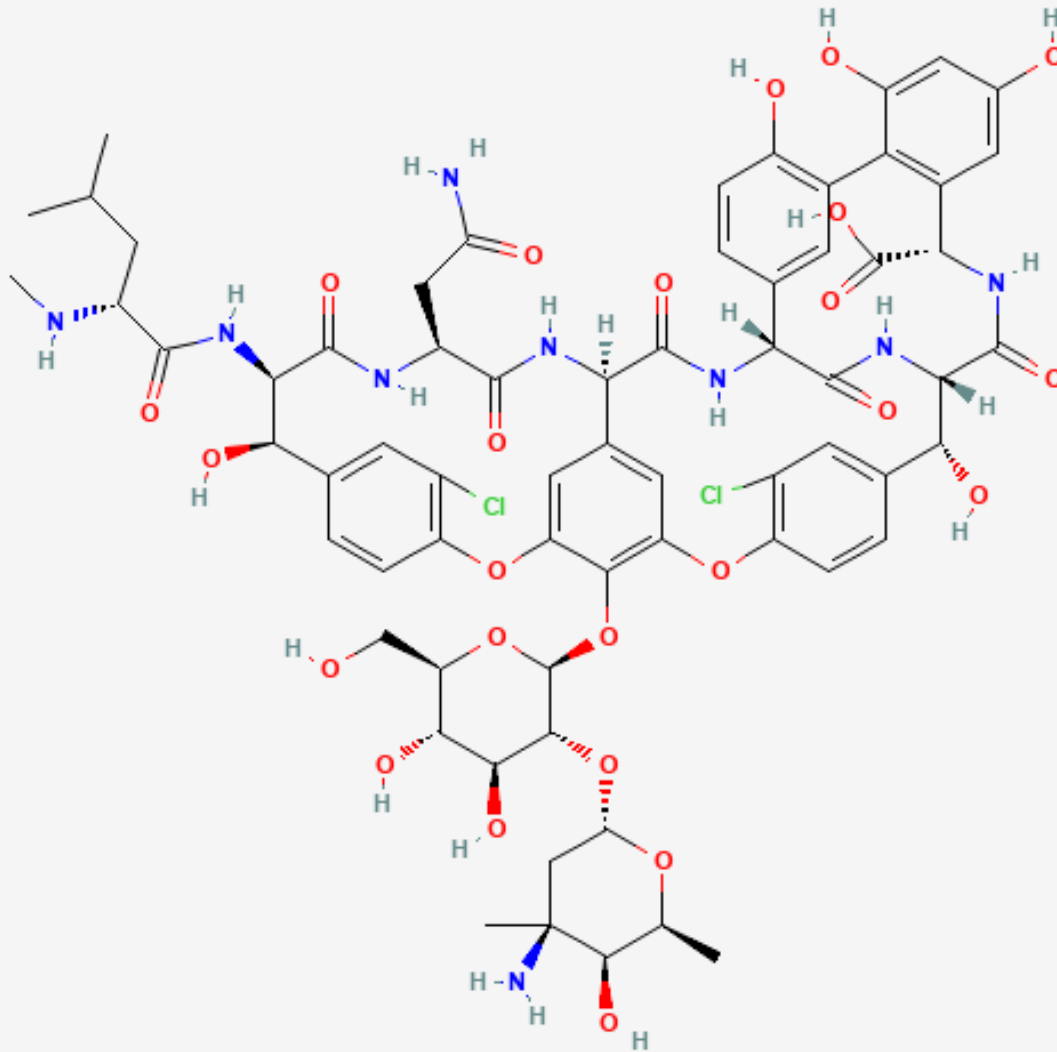
- \* Peptide hormone
- \* causes vasoconstriction
- \* leads to high B.P.
- \* B.P. medicines target this

$2^{11} = 2048$  possible isomers

1024 enantiomers

Evolution has forced  
formation of a single  
enantiomer

# Vancomycin

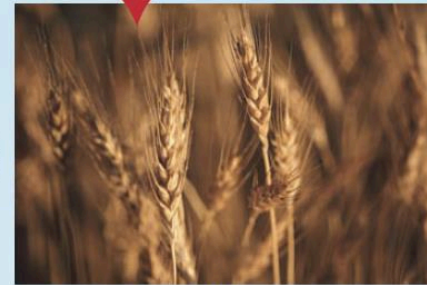


Glycopeptide  
antibiotic for  
gram (+) bacteria

18 chiral centres, 1 chiral plane, 1 chiral axis



foods rich in starch



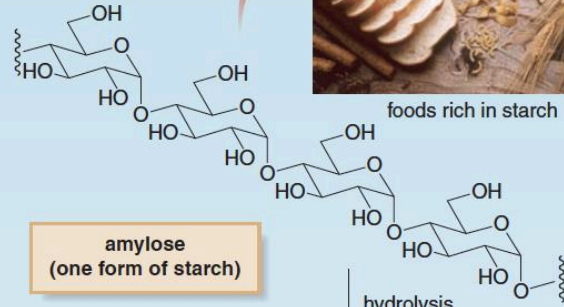
wheat



cotton plant

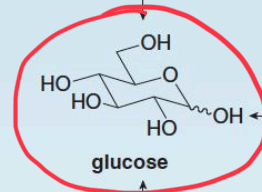


cotton fabric



amylose  
(one form of starch)

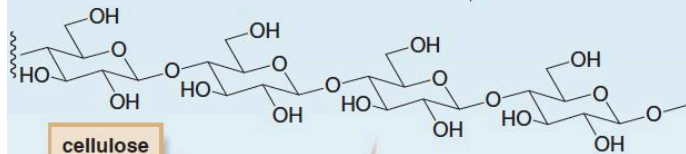
hydrolysis



glucose

This OH can be either  
axial or equatorial.

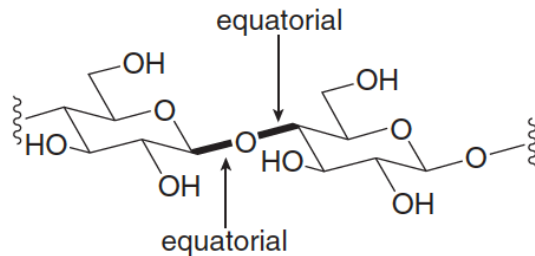
hydrolysis



cellulose

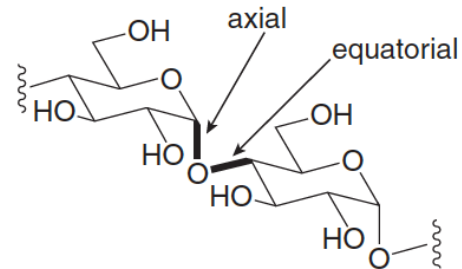
- In cellulose, the O atom joins two rings using two equatorial bonds.
- In starch, the O atom joins two rings using one equatorial and one axial bond.

**Cellulose**



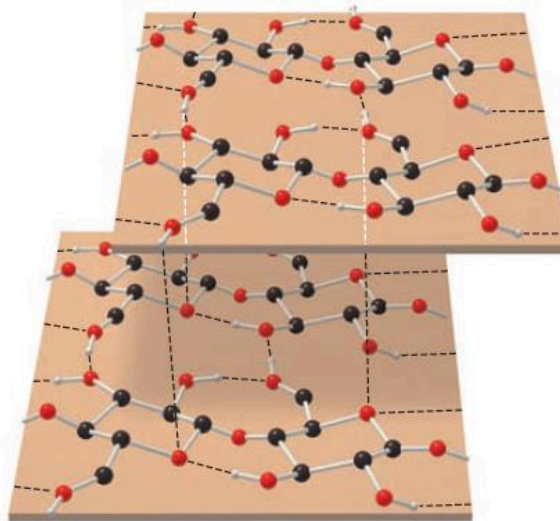
two equatorial bonds

**Starch**



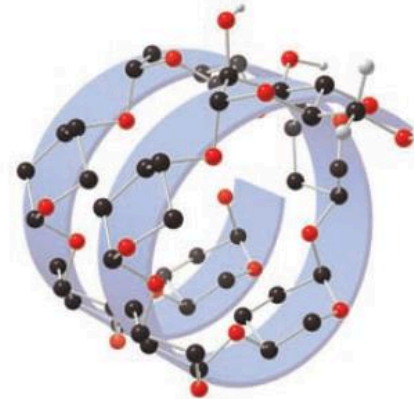
one axial, one equatorial bond

**Cellulose**



Cellulose consists of an extensive three-dimensional network held together by hydrogen bonds.

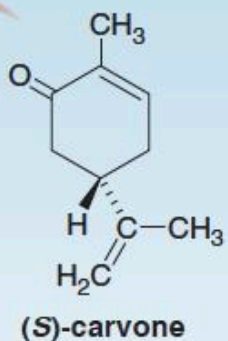
**Starch**



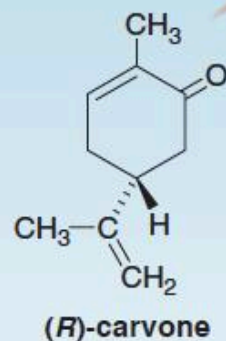
The starch polymer is composed of chains that wind into a helix.



caraway seeds



(S)-Carvone has the odor of caraway.



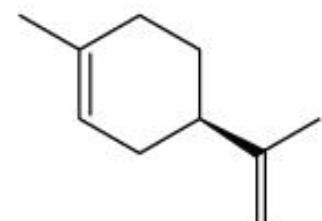
(R)-Carvone has the odor of spearmint.



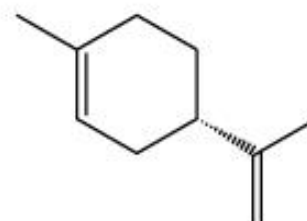
spearmint leaves



Media Bakery



**(+)-Limonene**  
(the enantiomer  
of limonene found  
in oranges)

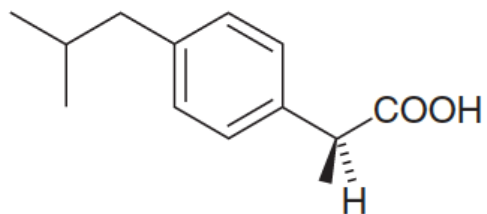


**(-)-Limonene**  
(the enantiomer  
of limonene found  
in lemons)



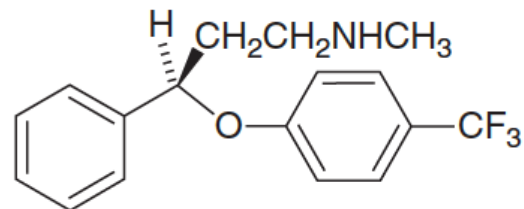
Media Bakery

# Chiral drugs



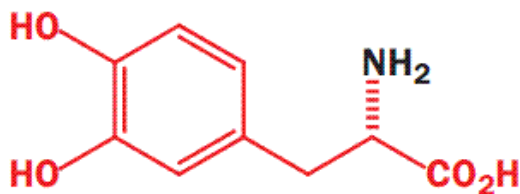
**(S)-ibuprofen**  
anti-inflammatory agent

Motrin/Advil



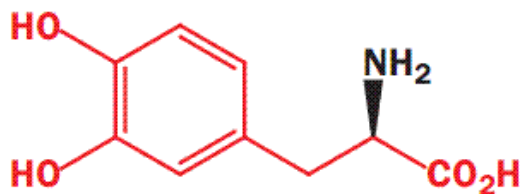
**(R)-fluoxetine**  
antidepressant

Prozac



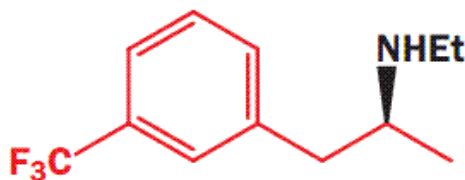
L-dopa  
marketed as a single enantiomer

Crosses blood-brain barrier

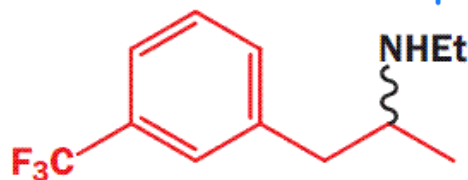


D-dopa  
toxic

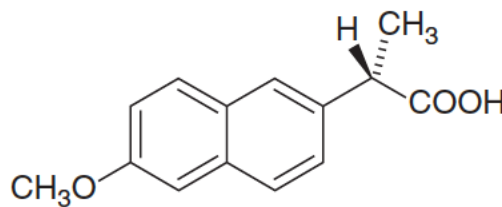
For treatment of  
Parkinson's disease  
precursor to  
neurotransmitter  
Dopamine



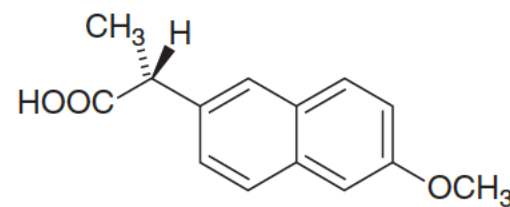
dexfenfluramine is an  
appetite suppressant



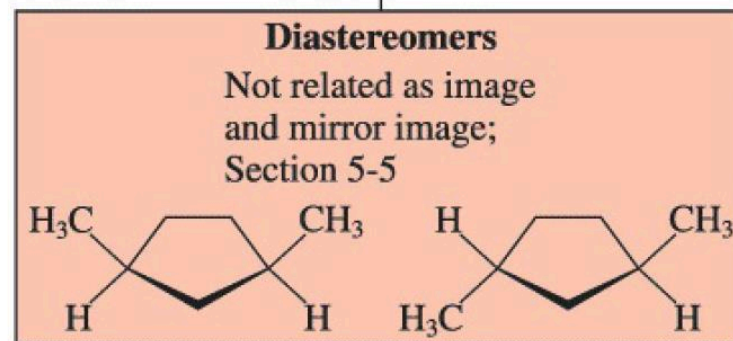
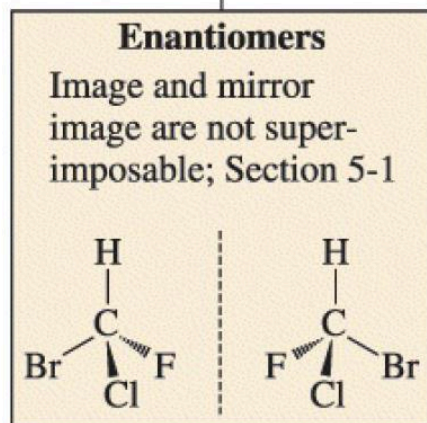
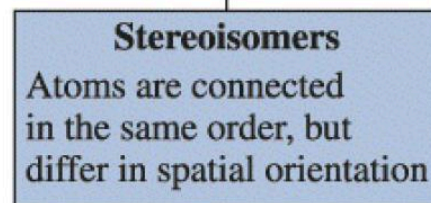
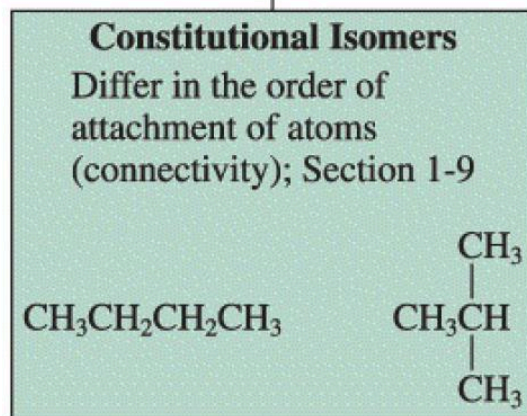
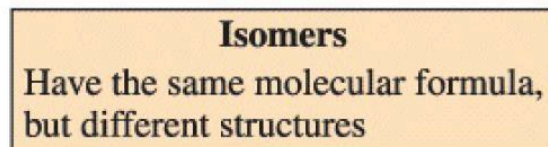
racemic fenfluramine has  
undesirable side-effects



(S)-naproxen  
anti-inflammatory agent



(R)-naproxen  
liver toxin



ISOMERS (same M.F. but different compounds)

Structural isomers

↓ Difference in atom-to-atom connectivity

$C_5H_{12}$

n-pentane  $CH_3CH_2CH_2CH_2CH_3$

iso-pentane  $\text{>CH-CH}_2\text{CH}_3$

neo-pentane  $\begin{array}{c} H_3C \\ | \\ C-CH_3 \\ | \\ H_3C \end{array}$

Spatial isomers (stereoisomers)

↓ difference in spatial arrangement of groups/atoms w.r.t. each other

optical

Geometrical

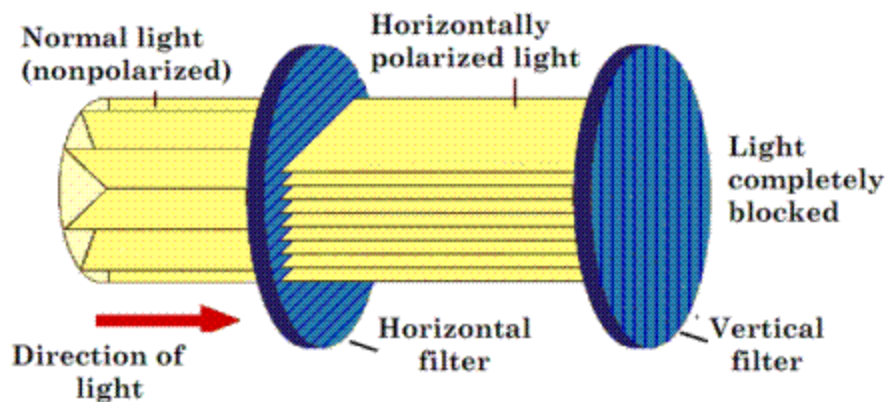
Conformational isomers:  
difference in torsion angles



Dutch  
astronomer  
physicist  
mathematician  
(1629-1695)

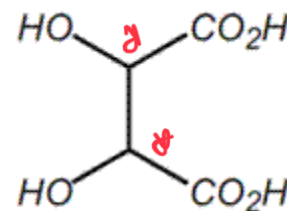
Christian Huygens

## PLANE POLARIZED LIGHT



1742-1786

Carl Wilhelm Scheele



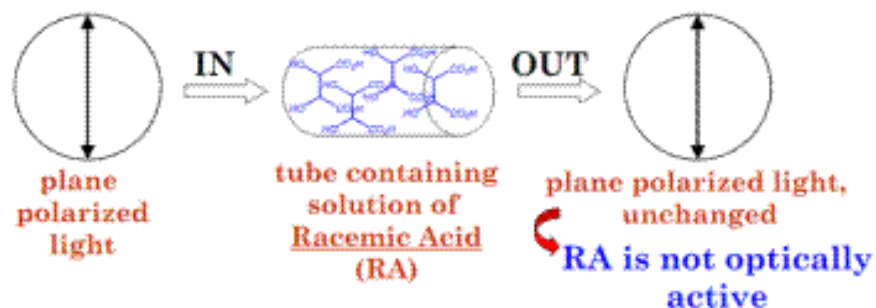
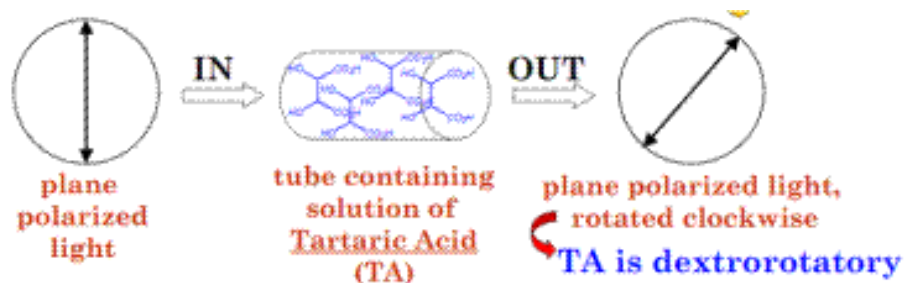
Tartaric Acid

barrels of grape fermentation

History of Stereochemistry

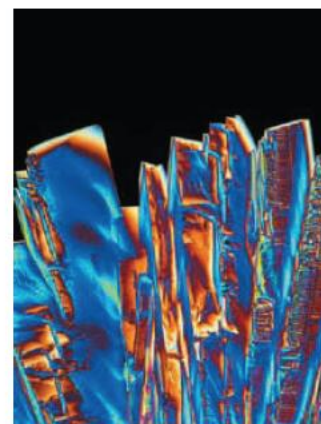


Jean Baptiste Biot.

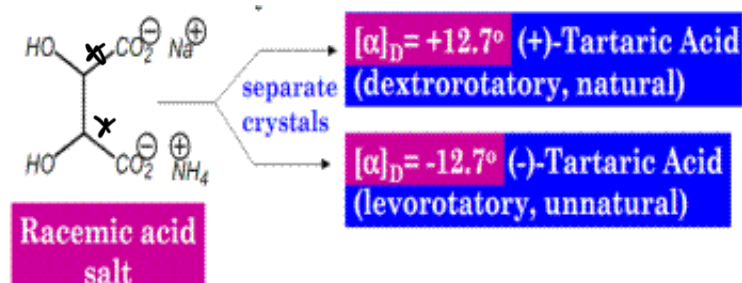


Louis Pasteur

handpicked  
mirror  
image  
crystals of  
sodium  
ammonium  
tartarate

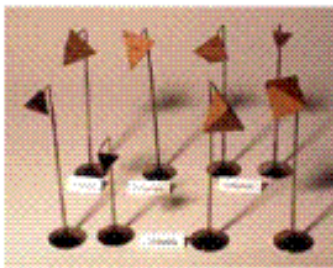


Tartaric acid crystals.



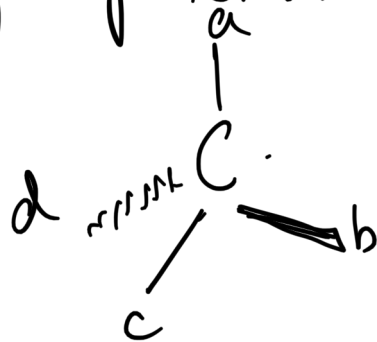


**Joseph A.  
LeBel**  
(1847-1930)



**Jacobus H.  
van't Hoff**  
(1852-1930)

Concept of tetrahedral carbon



asymmetric  
carbon

Stereogenic  
centre.



## EMIL FISHER



- Sugar Chemist
- stereochemical configuration of 16 aldohexoses in 1894
- Fisher Projection formulae (2D representation)

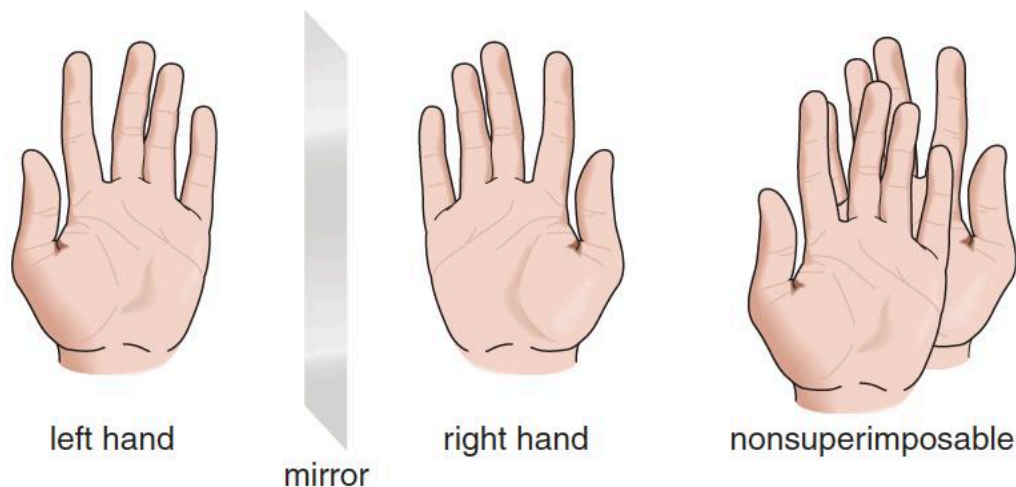


## VLADIMIR PRELOG

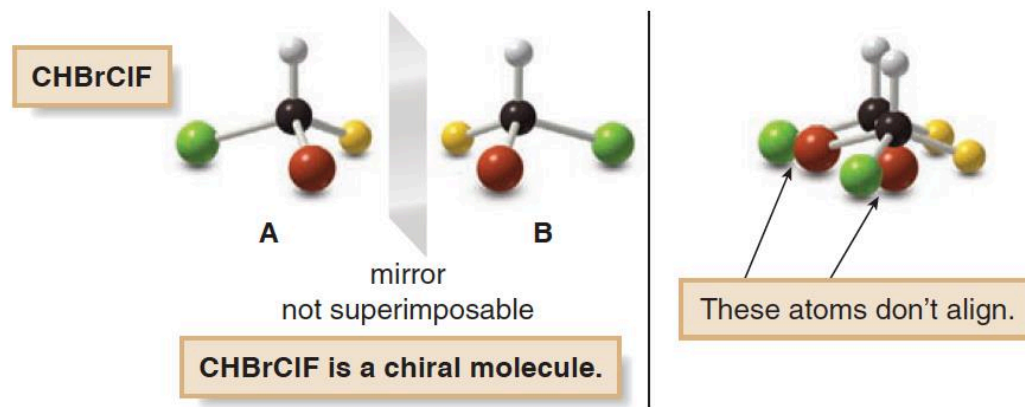
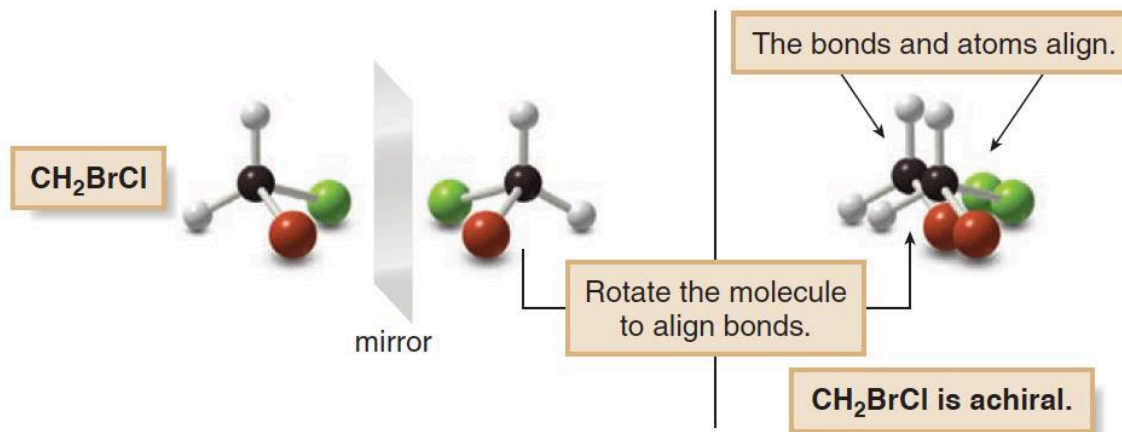
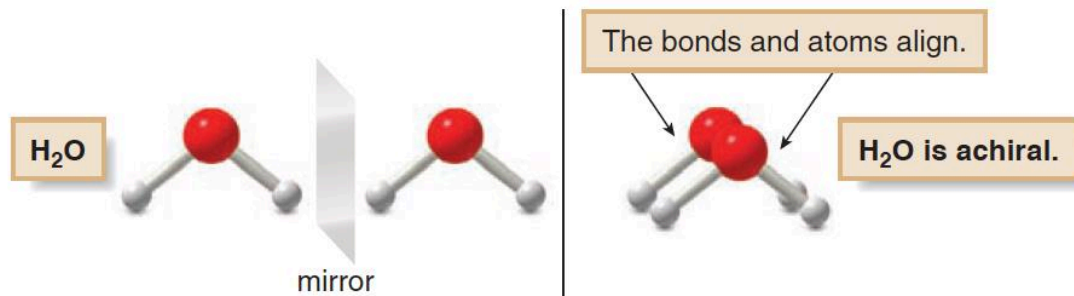
- 1966 R/S & E/Z
- 1975 Nobel Prize in Chemistry

## Chiral and Achiral Molecules

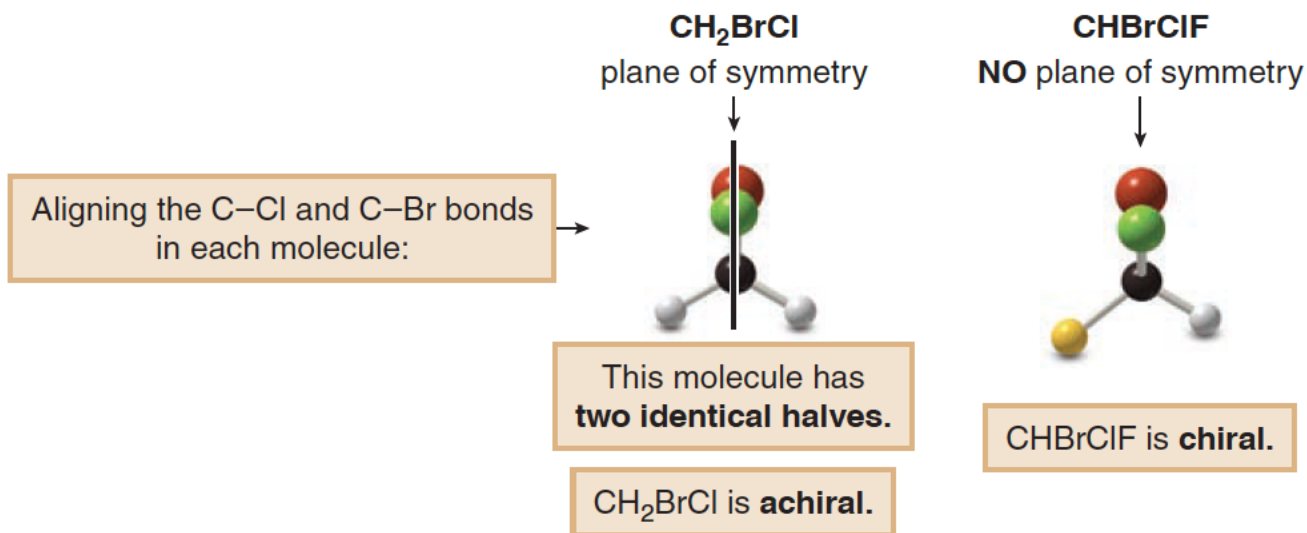
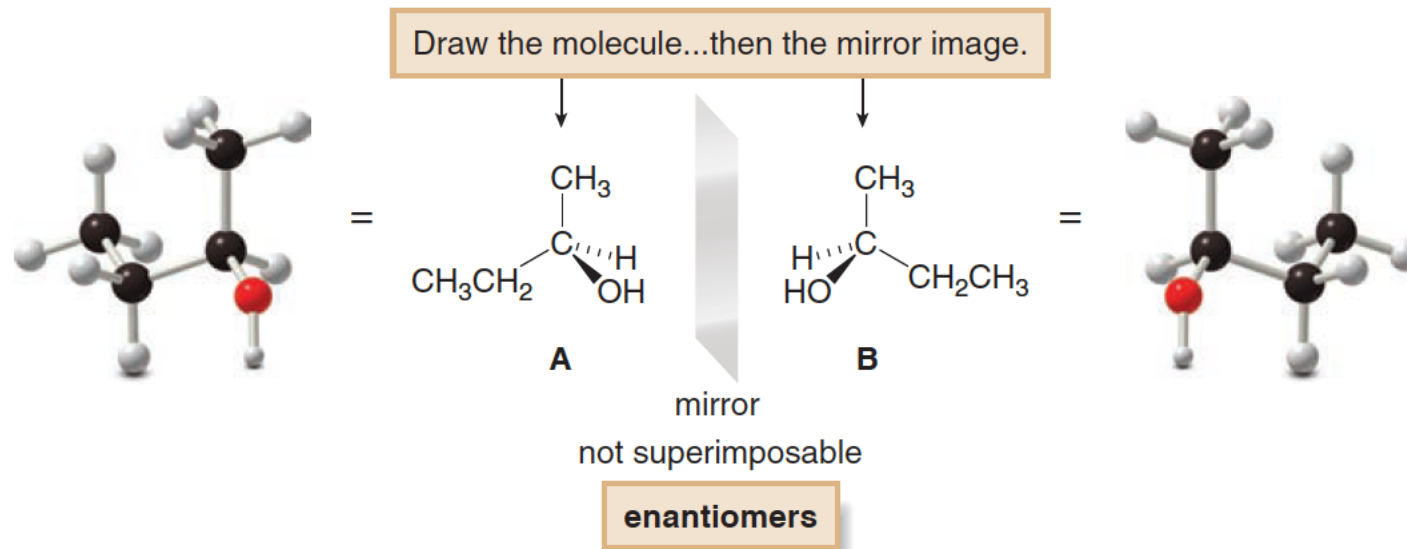
Chiral and Achiral Molecules  
↓  
asymmetric      ↓  
symmetric



A molecule (or an object) that is non-superimposable on its mirror image is said to be chiral

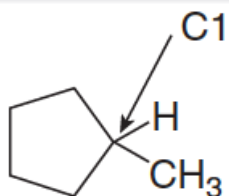


Non-superimposable mirror images like A and B are called enantiomers

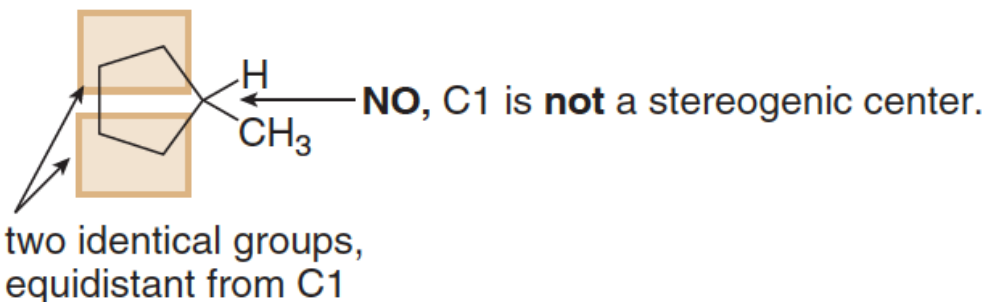


# Stereogenic Centres in a ring

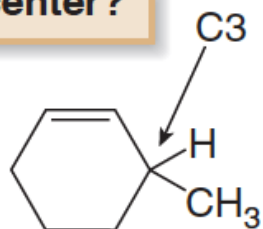
Is C1 a stereogenic center?



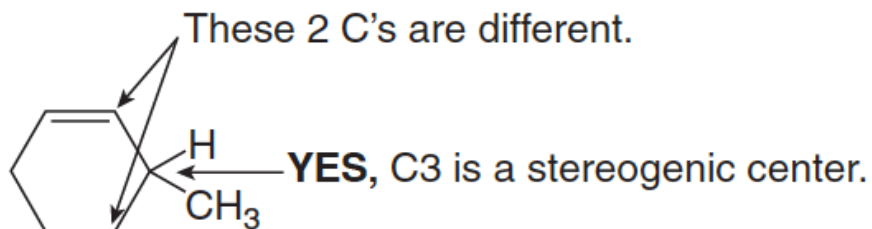
methylcyclopentane



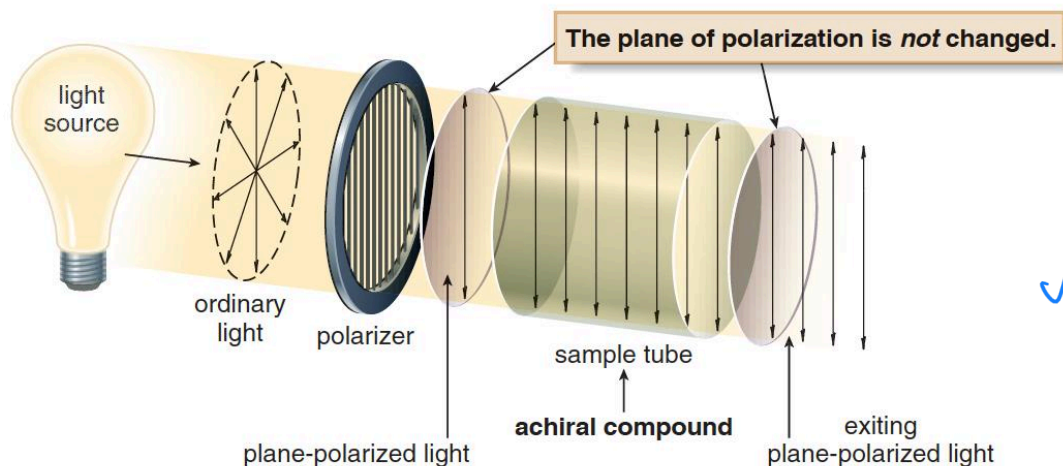
Is C3 a stereogenic center?



3-methylcyclohexene



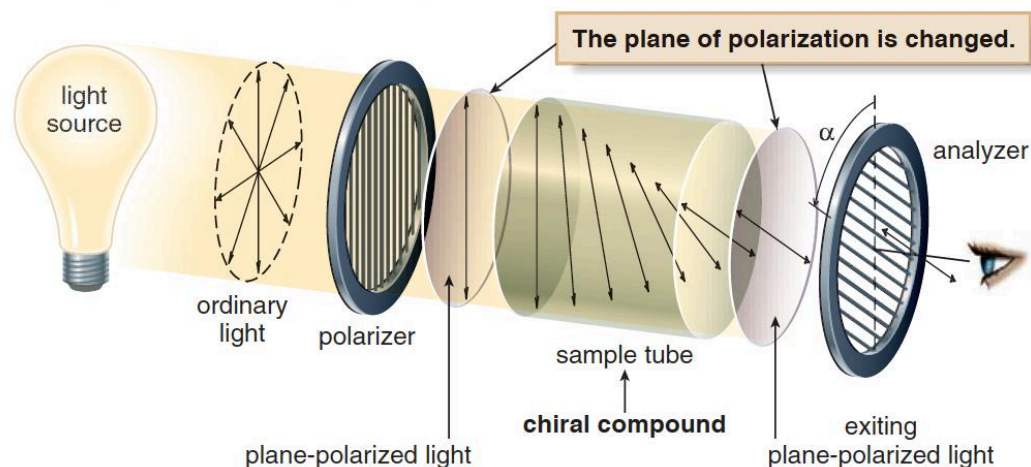
# Optical activity



Achiral

OPTICAL  
ACTIVITY  
polarimeter

With **chiral compounds**, the plane of the polarized light is rotated through an angle  $\alpha$ . The angle  $\alpha$ , measured in degrees ( $^\circ$ ), is called the observed rotation. A compound that rotates the plane of polarized light is said to be **optically active**.



Chiral

clockwise rotation  
(+), dextrorotatory (d)

anticlockwise  
(-), levorotatory (l)

$$\text{specific rotation} = [\alpha]_{\lambda}^T = \frac{\alpha}{l \times c}$$

$\alpha$  = observed rotation ( $^{\circ}$ )  
 $l$  = length of sample tube (dm)  
 $c$  = concentration (g/mL)

[ dm = decimeter  
 1 dm = 10 cm ]

Specific rotation is a standardized physical constant for the amount that a chiral compound rotates plane-polarized light. Specific rotation is denoted by the symbol  $[\alpha]$  and defined using a specific sample tube length ( $l$  in dm), concentration ( $c$  in gm/mL), temperature ( $25^{\circ}\text{C}$ ) and wavelength (589 nm) which is yellow light from sodium lamp.

No. of degrees of rotation when the conc. of sample is fixed at 1g/mL & length of tube 1dm (10cm)

Degree of rotation ( $\alpha$ )  $\rightarrow$

1. Temp.  $T \uparrow \Rightarrow \alpha \uparrow$
2.  $\lambda \quad \alpha \propto \frac{1}{\lambda^2}$
3.  $\alpha \propto [c]$
4.  $\alpha \propto l$
5. Solvent

## Enantiomeric excess and Optical purity: ee and op

- **Enantiomeric excess (ee)** is a measurement of the excess of one enantiomer over the racemic mixture.

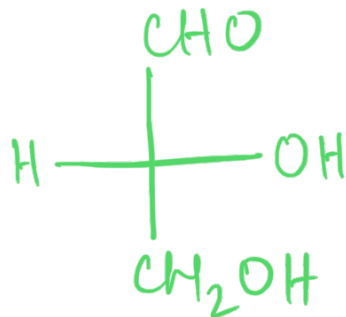
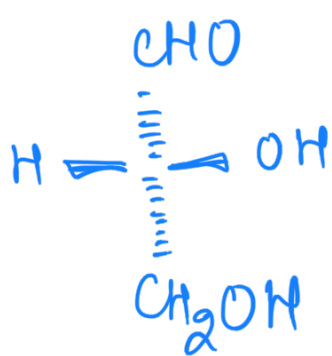
$$ee = \% \text{ of one enantiomer} - \% \text{ of the other enantiomer.}$$

- Consider the following example: If a mixture contains 75% of one enantiomer and 25% of the other, the enantiomeric excess is  $75\% - 25\% = 50\%$ . Thus, there is a 50% excess of one enantiomer over the racemic mixture.
- ee is numerically equal to **Optical Purity**.
- The optical purity can be calculated if the specific rotation  $[\alpha]$  of a mixture and the specific rotation  $[\alpha]$  of a pure enantiomer are known.

$$op = ([\alpha] \text{ mixture} / [\alpha] \text{ pure enantiomer}) \times 100.$$

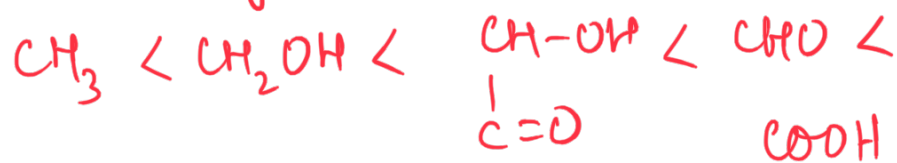
# FISHER PROTECTION

2D-representation of a 3D-molecule



Glyceraldehyde

Order of oxidation states



- always an eclipsed conformation
- place the longest carbon chain vertically
- highest oxidized group on top
- main chain bends away
- substituents stick out towards you

## optical rotation

(S)(+)-2-bromobutane

$$[\alpha]_D^{21} = +23.1^\circ \text{ for pure (S) - isomer}$$

$$[\alpha]_D^{21} = +9.2^\circ \text{ for the mixture}$$

What is the % of R & S in the mixture? What is ee & op?

$$ee = op = \frac{[\alpha]_{\text{mix}}}{[\alpha]_{\text{pure}}} \times 100 = \frac{+9.2}{+23.1} \times 100 = 40\%$$

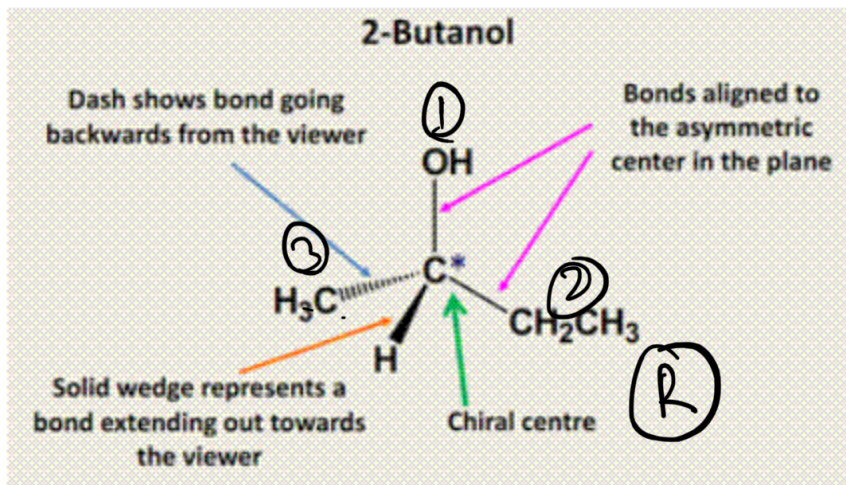
$$\begin{aligned} ee &= 40\% S + 60\% (R \& S) \\ &= 40\% S + 30\% S + 30\% R \\ &= 70\% S + 30\% R \end{aligned}$$

$$op = 70\%$$

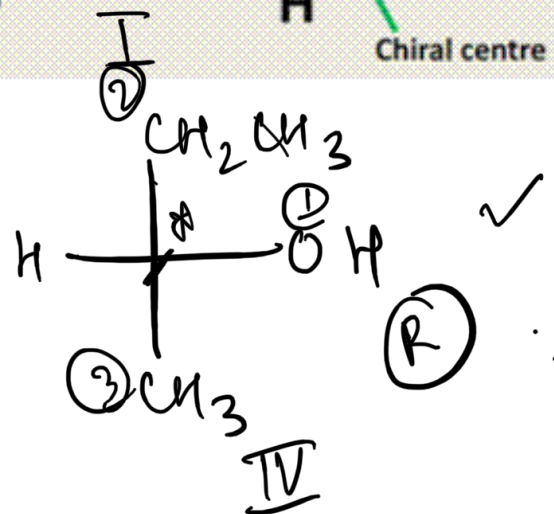
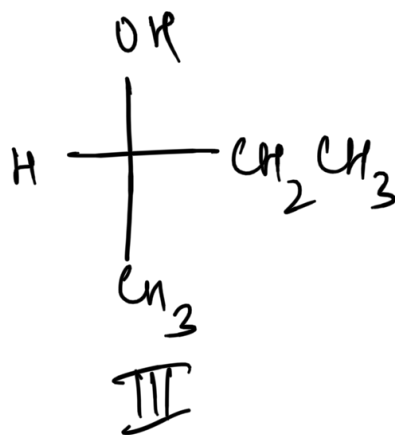
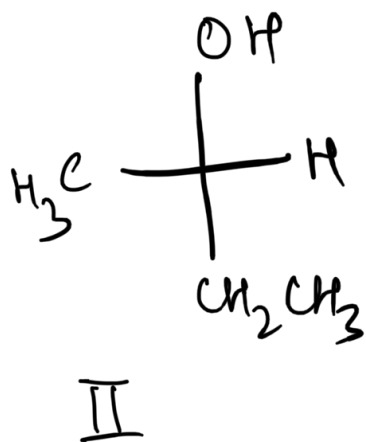
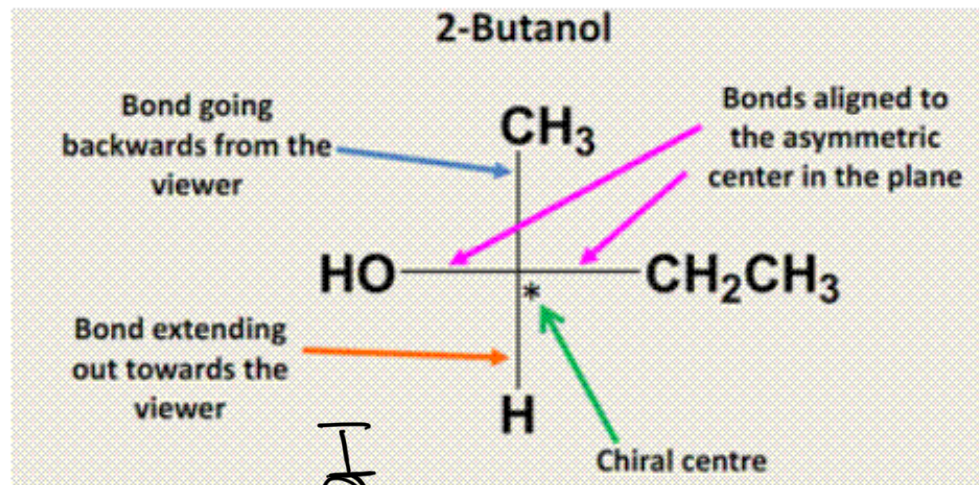
$$\text{Molar rotation} = \frac{\text{sp. rotation} \times \text{MW}}{100}$$

To take into consideration the molecular weight of compounds into observed optical rotation.

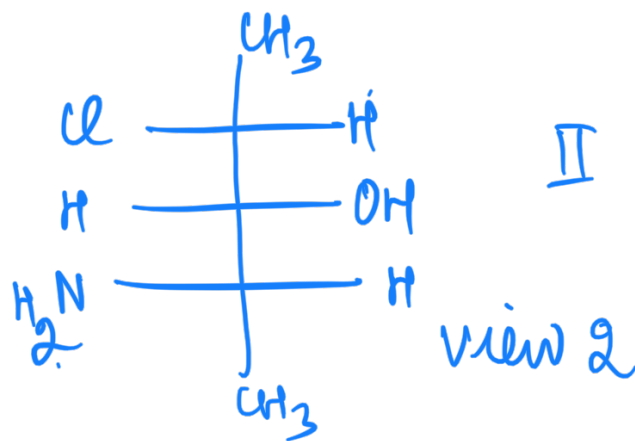
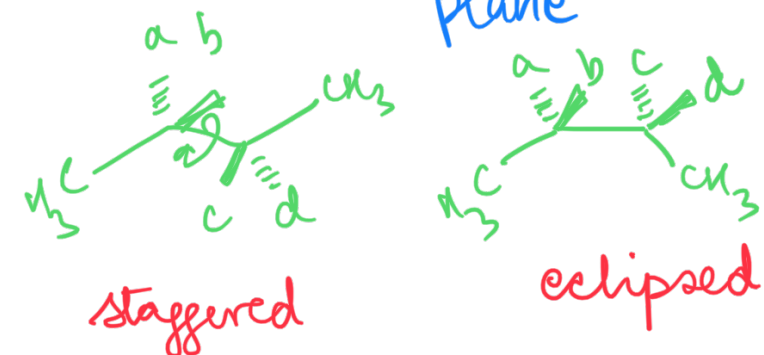
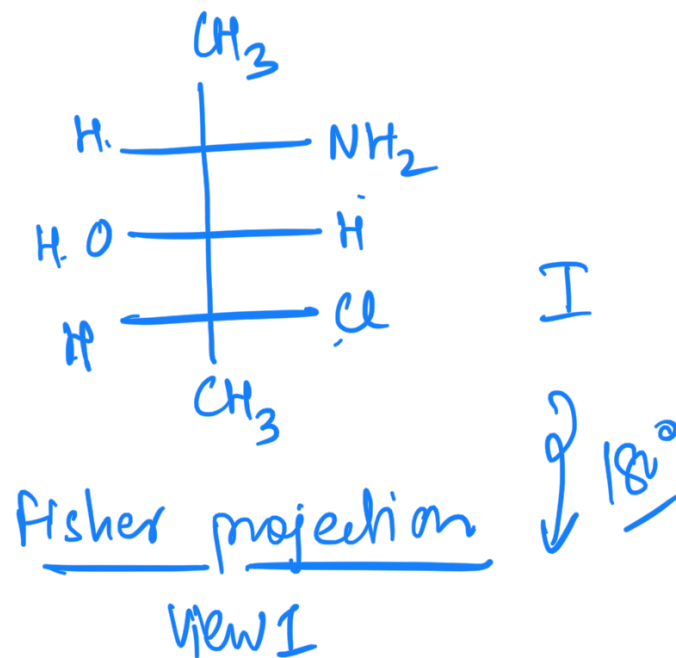
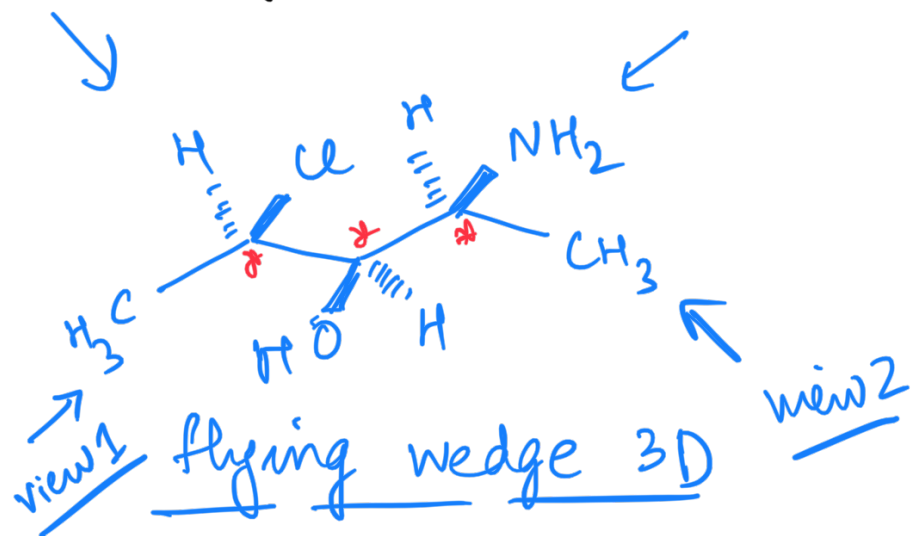
## 3D- flying wedge



## 2D- representation



# Zig-zag structure (many stereocentres)



# Stereochemical descriptors

Relative

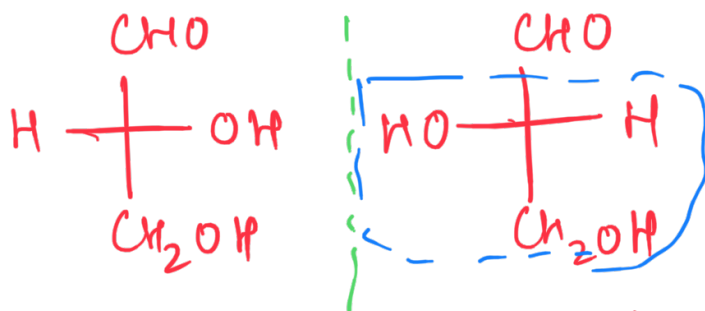
Absolute

D/L, erythro/threo

R/S

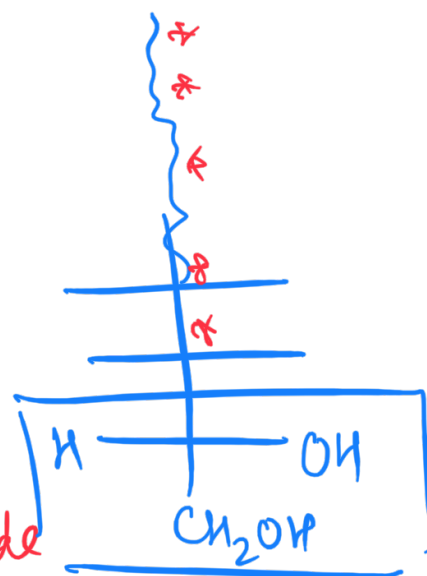
syn/anti endo/exo

cis/trans, E/Z



D(+) glyceraldehyde

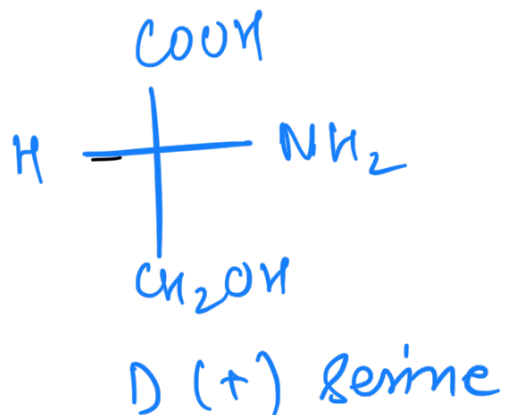
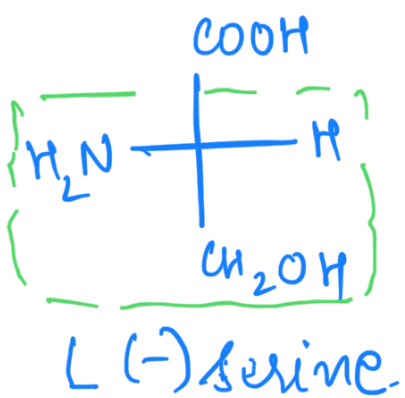
L(-) glyceraldehyde



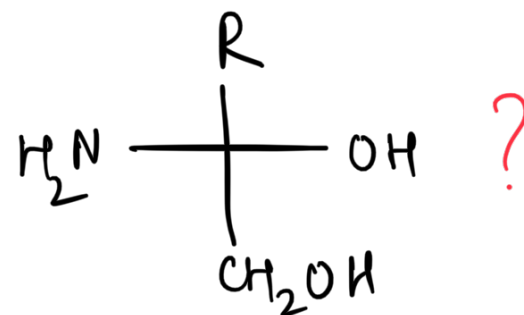
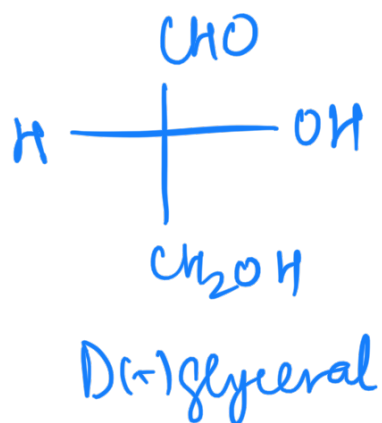
D -  
Sugar

D  $\neq$  d/(+)/dextro

L  $\neq$  l/(-)/levo



D (e) -  
 L (d) +  
 D (d) +  
 L (e) -



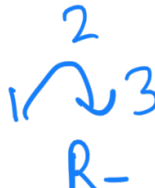
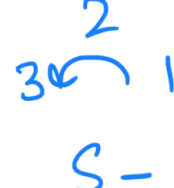
Ambiguity

D<sub>g</sub>

L<sub>s</sub>

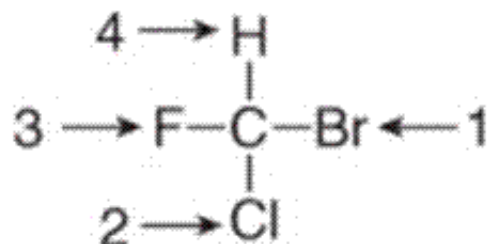
Need for abs. config

# ABSOLUTE CONFIGURATION (R/S)

- Precise arrangement of groups/atoms around a stereogenic carbon
- Cahn Ingold Prelog 1951 (CIP)
  - \* Assign priorities to all groups/atoms around tetrahedral carbon : Sequence Rules
  - \* Lowest priority group away from the viewer
  - \* Trace a circle 1, 2, 3
    -  R-
    -  S-

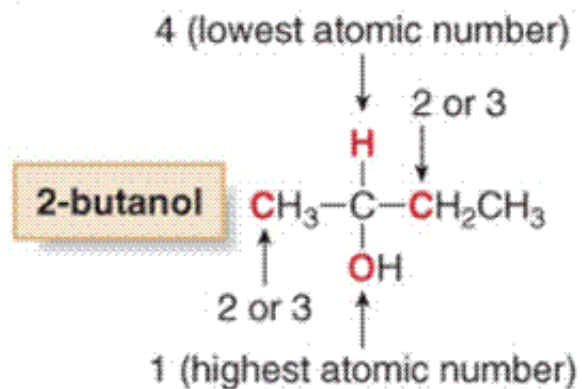
## CIP Sequence Rules:

- The atom of **highest atomic number** gets the highest priority .

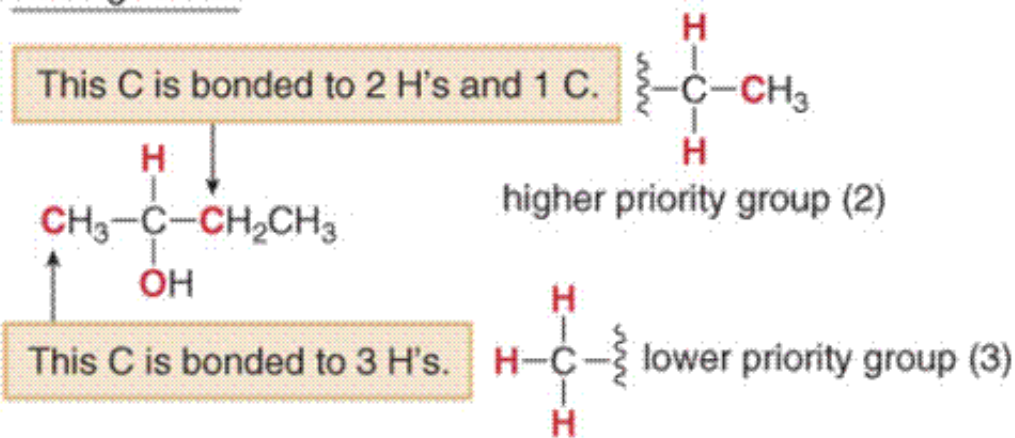


- If priority cannot be decided based on the first atom attached, assign priority based on the atomic number of the atoms bonded to these atoms.

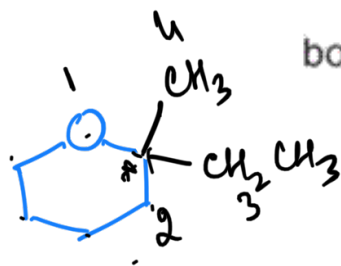
Following rule 1:



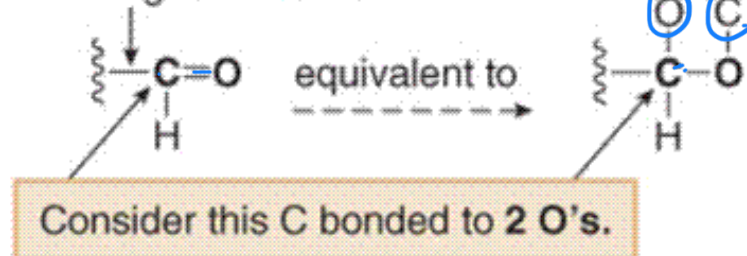
Adding rule 2:



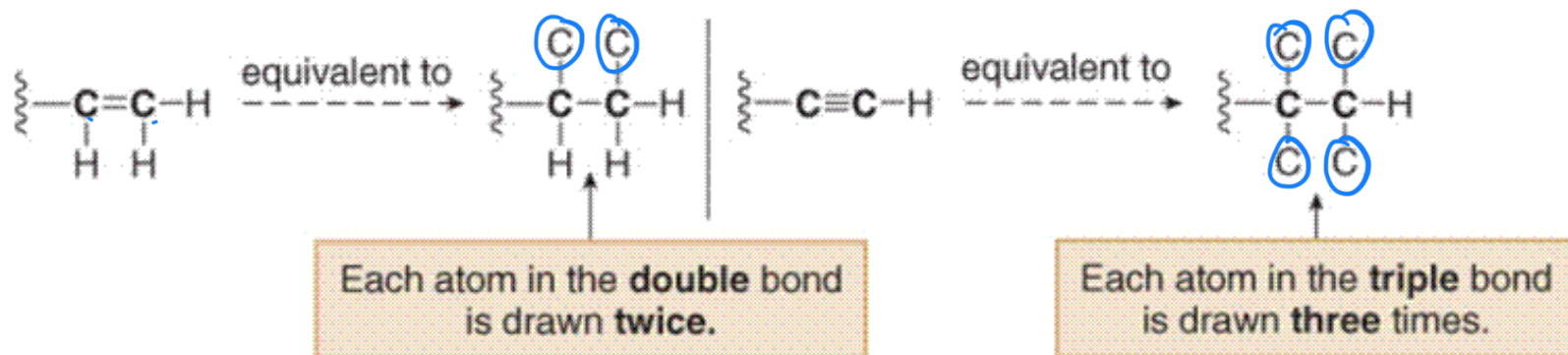
- If two isotopes are bonded to the stereogenic center, assign priorities in order of decreasing **mass number**. Thus, in comparing the three isotopes of hydrogen, the order of priorities is: tritium (1), deuterium (2), hydrogen (3)
- If the atom is a part of a multiple bond, treat the multiply bonded atom as an equivalent number of singly bonded atoms.



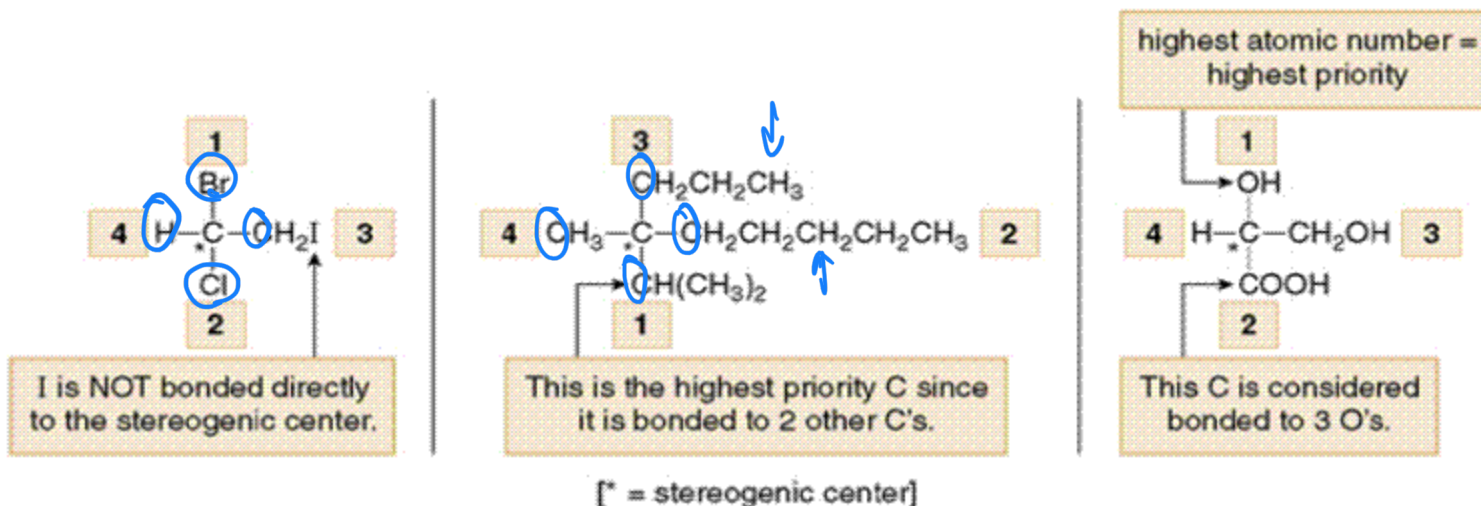
bonded to a stereogenic center here



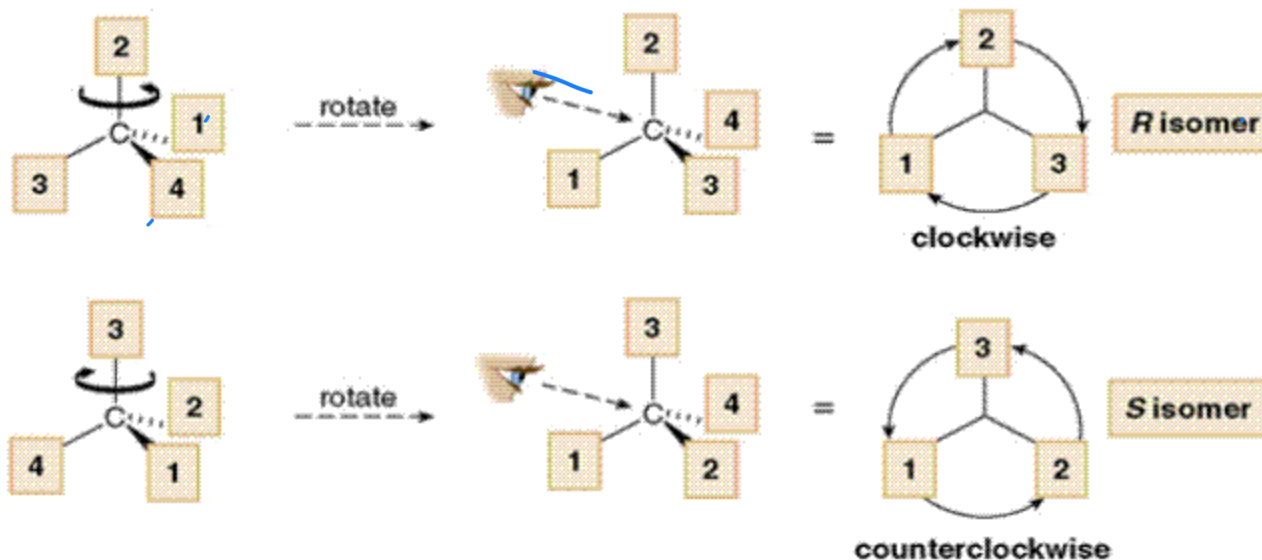
Phantom atoms

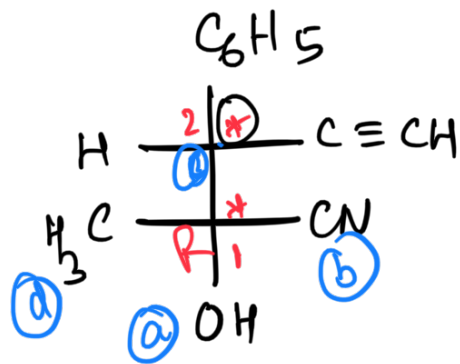
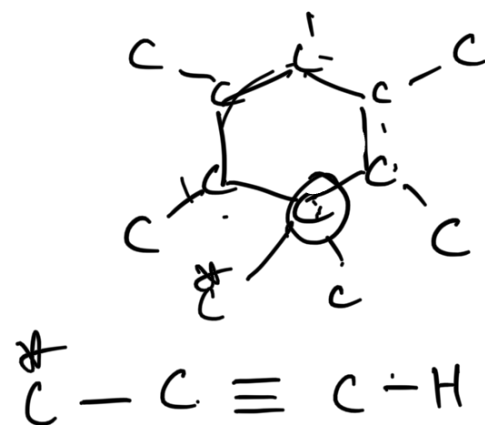
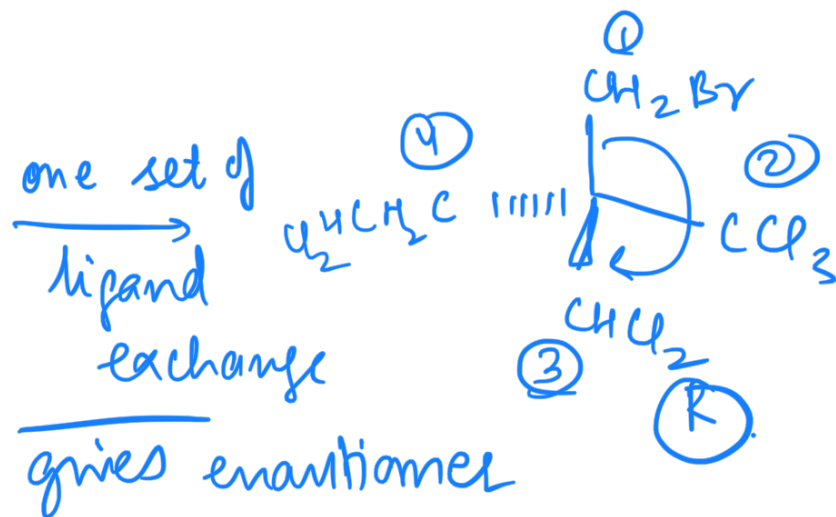
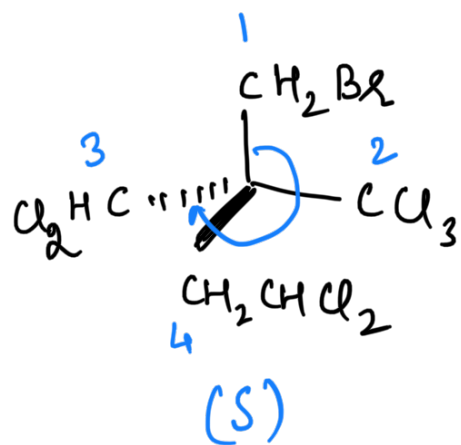


## Assigning priorities to stereogenic centers

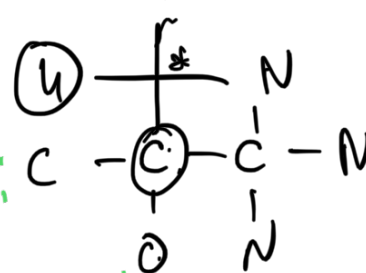
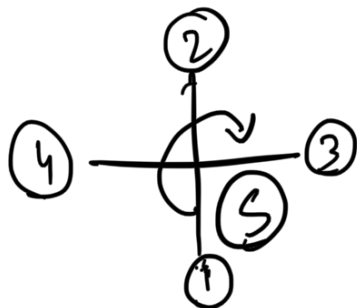
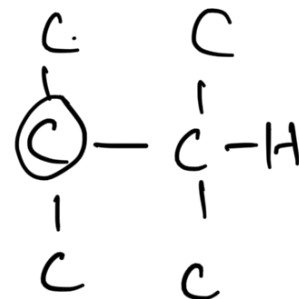
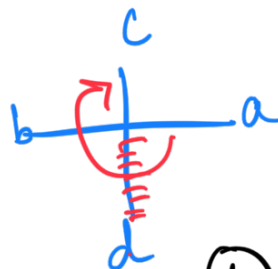


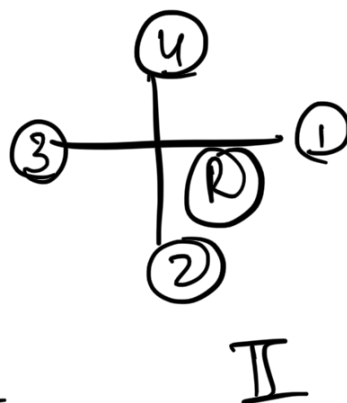
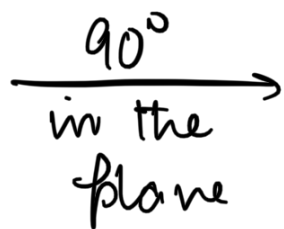
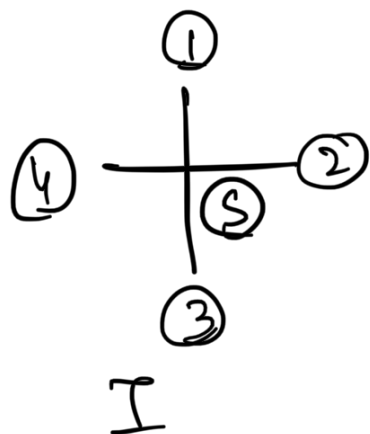
## Orienting the lowest priority group in back



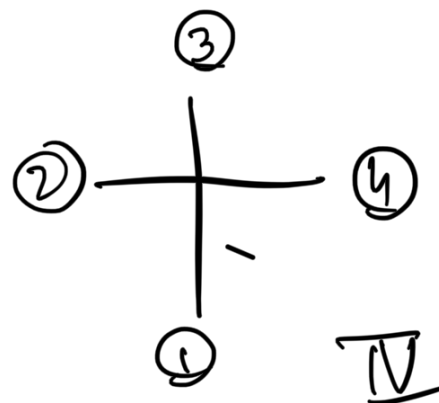
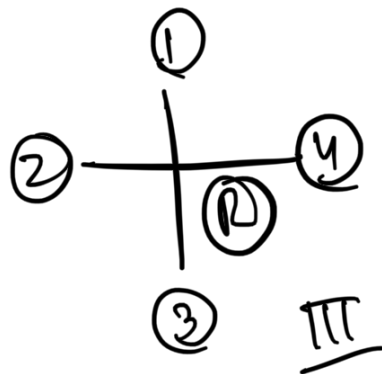
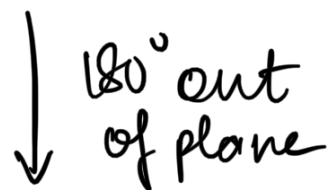


$\equiv$



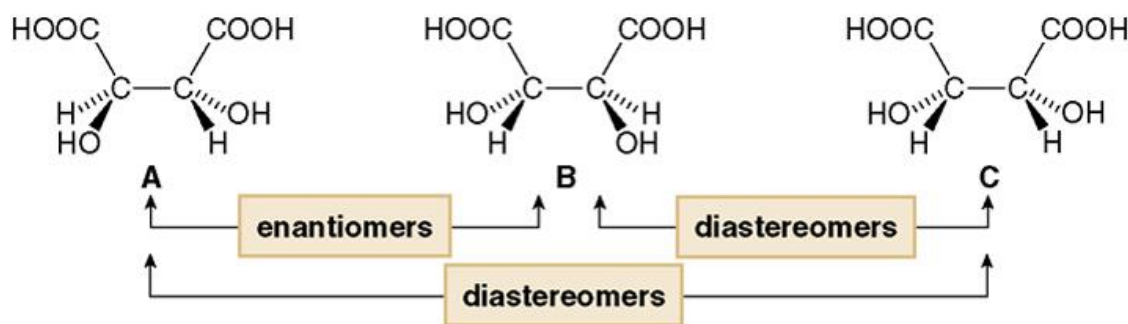


enantiomer



I & IV identical

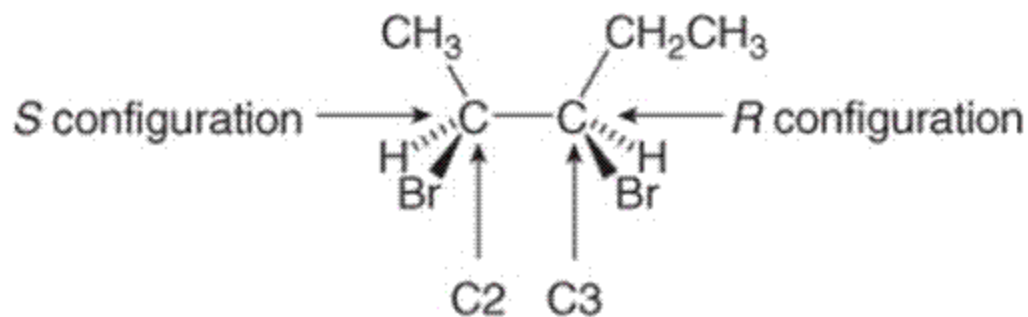
- Stereoisomers that are not mirror images of one another are called **diastereomers**.
- Enantiomers have identical physical properties and cannot be separated by common physical techniques like distillation.
- Diastereomers and constitutional isomers have different physical and chemical properties**, and therefore can be separated by common physical techniques.



| Property                               | A          | B          | C          | A + B (1:1) |
|----------------------------------------|------------|------------|------------|-------------|
| melting point (°C)                     | 171        | 171        | 146        | 206         |
| solubility (g/100 mL H <sub>2</sub> O) | 139        | 139        | 125        | 139         |
| [ $\alpha$ ]                           | +13        | -13        | 0          | 0           |
| <i>R,S</i> designation                 | <i>R,R</i> | <i>S,S</i> | <i>R,S</i> | —           |
| <i>d,l</i> designation                 | <i>d</i>   | <i>l</i>   | none       | <i>d,l</i>  |

- The physical properties of **A** and **B** differ from their diastereomer **C**.
- The physical properties of a racemic mixture of **A** and **B** (last column) can also differ from either enantiomer and diastereomer **C**.
- C** is an achiral meso compound, so it is optically inactive; [ $\alpha$ ] = 0.

- Identical compounds have the *same* *R,S* designations at every tetrahedral stereogenic center.
- Enantiomers have exactly *opposite* *R,S* designations.
- Diastereomers have the *same* *R,S* designation for at least one stereogenic center and the *opposite* for at least one of the other stereogenic centers.



(2*S*,3*R*)-2,3-dibromopentane

# Molecules with multiple stereocenters

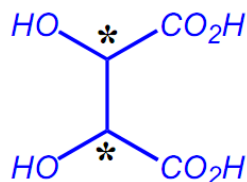
Molecules with **1 stereocenter** can be R or S

→ 2 possible stereoisomers

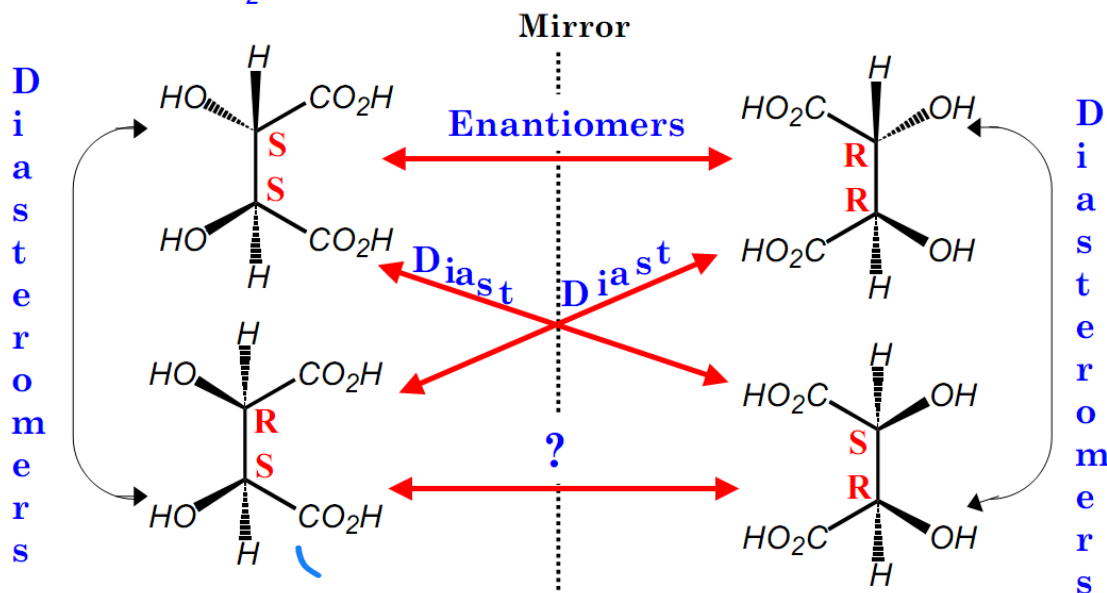
Molecules with  **$n$  stereocenters** can have all the possible combinations of R and S for each stereocenter

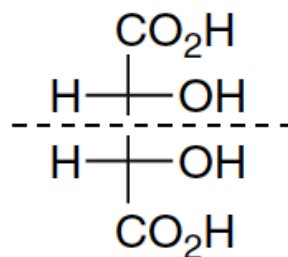
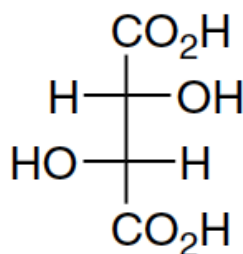
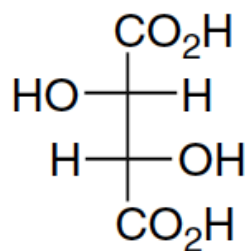
→  $2^n$  possible stereoisomers

## Tartaric Acid



2 stereocenters → 4 possible stereoisomers

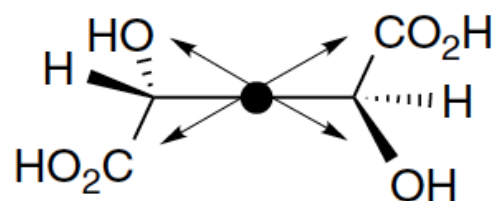
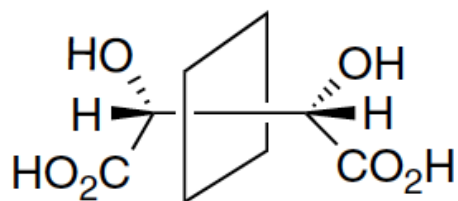




D-tartaric acid

L-tartaric acid

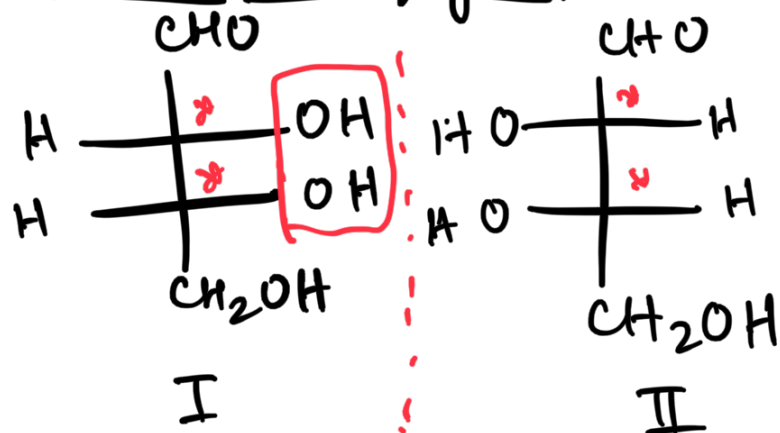
*meso*-tartaric acid



Plane of symmetry in the eclipsed conformation of *meso*-tartaric acid

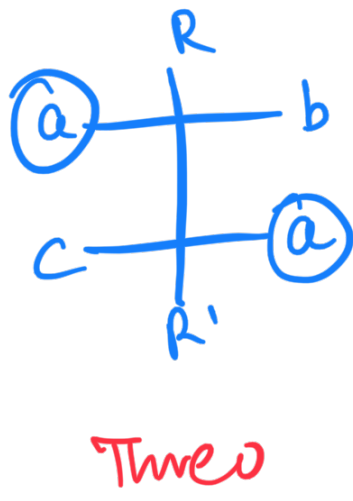
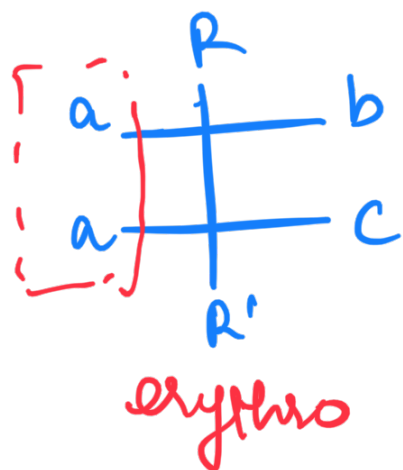
Center of symmetry in the *anti* staggered conformation of *meso*-tartaric acid

# Relative configuration

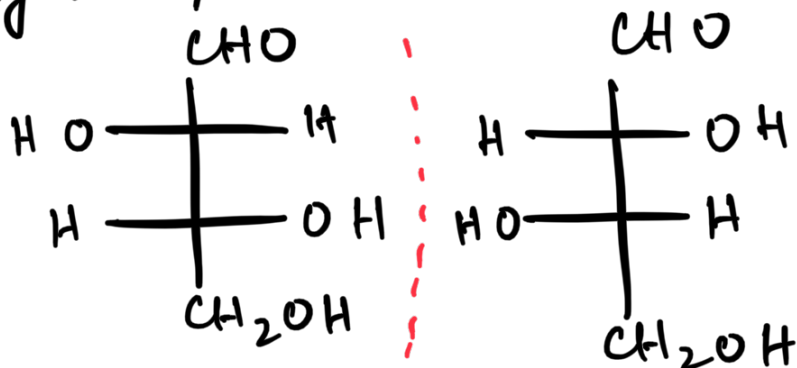


(-) erythrose

(+) erythrose



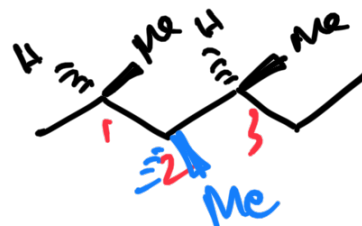
## erythro / threo



(-) Threose

(+) Threose

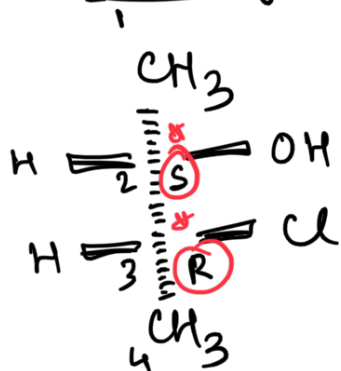
If same sub. are on same side eclipsing → erythro  
opp. side → threo



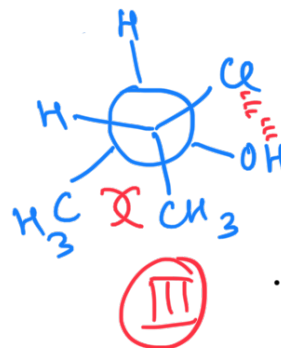
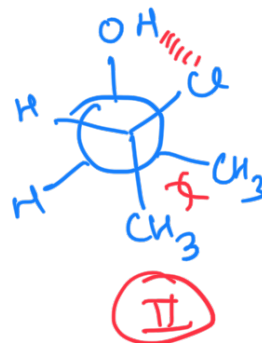
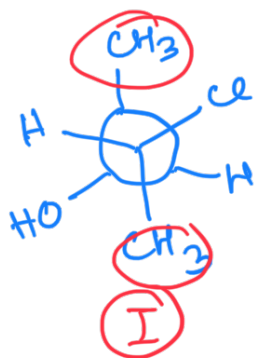
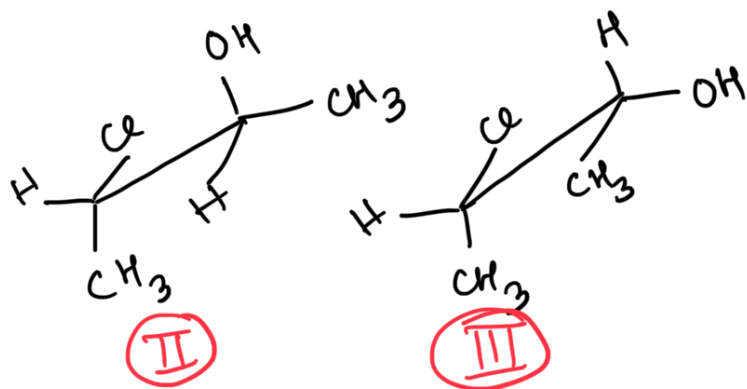
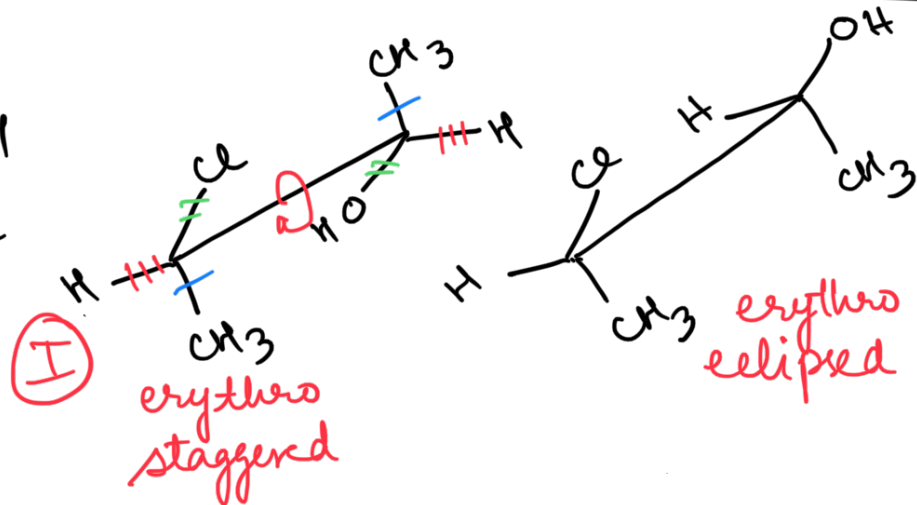
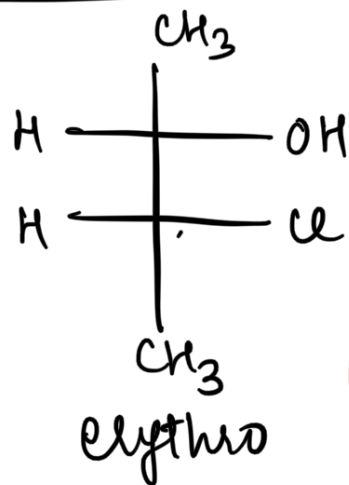
1,3-syn

1,2-syn

# Transformation from Fischer $\rightarrow$ Sawhorse $\rightarrow$ Newmann



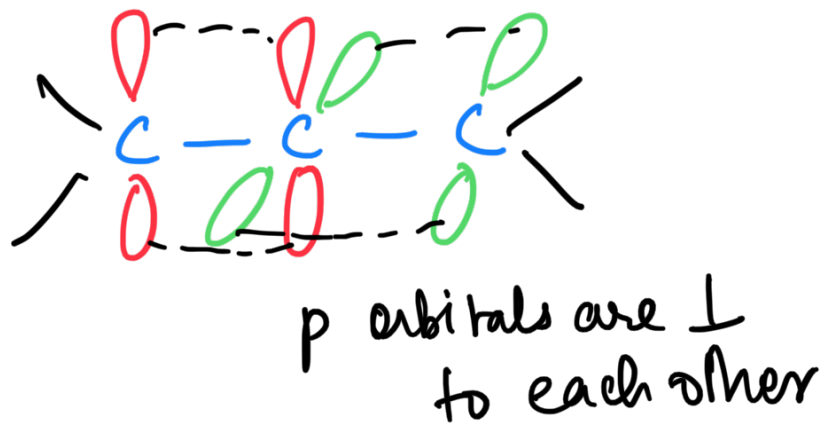
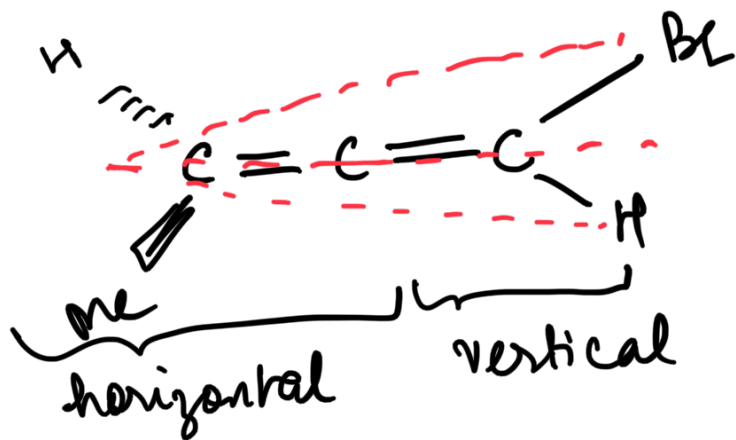
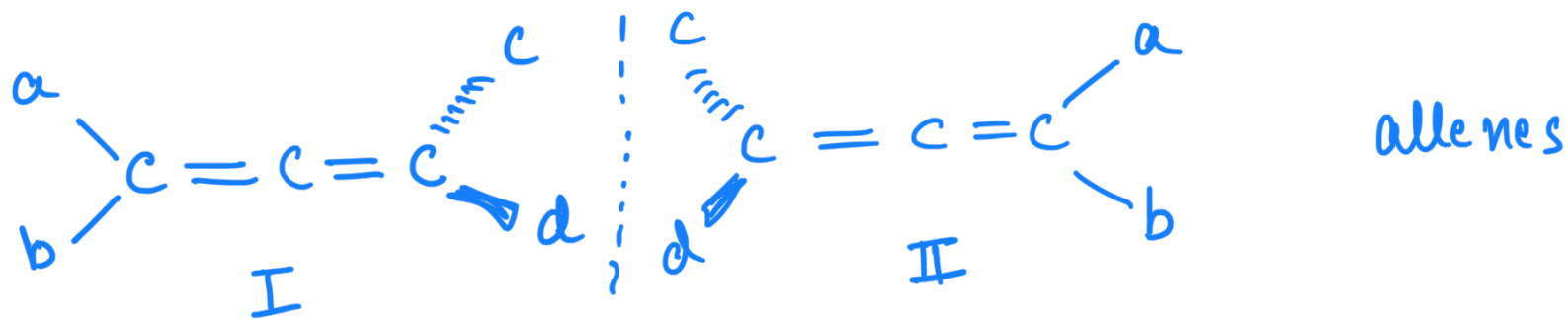
(3R)-chloro-  
(2S)-butanol



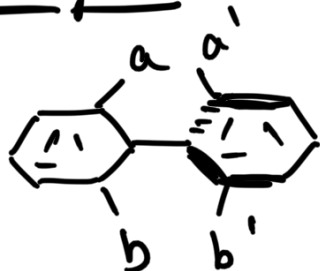
$II \sim III > I$

$II \sim III > I$

Axial chirality: molecule does not possess any stereogenic carbon but it has a chiral axis around which there is a non-planar arrangement of atoms or groups.



# biphenyls



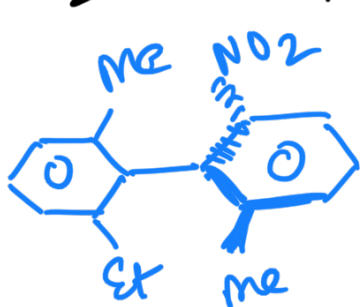
$$a \neq b$$
$$a' \neq b'$$

Restricted rotation around C-C  $\sigma$  bond  
ATROP ISOMERISM  $\rightarrow$  axial chirality

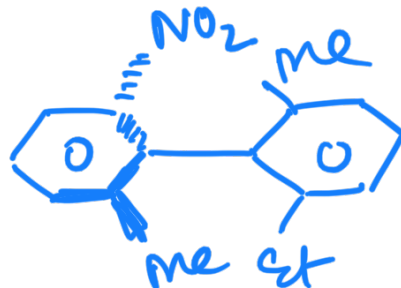
Order of bulky gps



Requisite : 1. presence of bulky substituents on ortho-C's.  
2.  $a \neq b$ ,  $a' \neq b'$



I

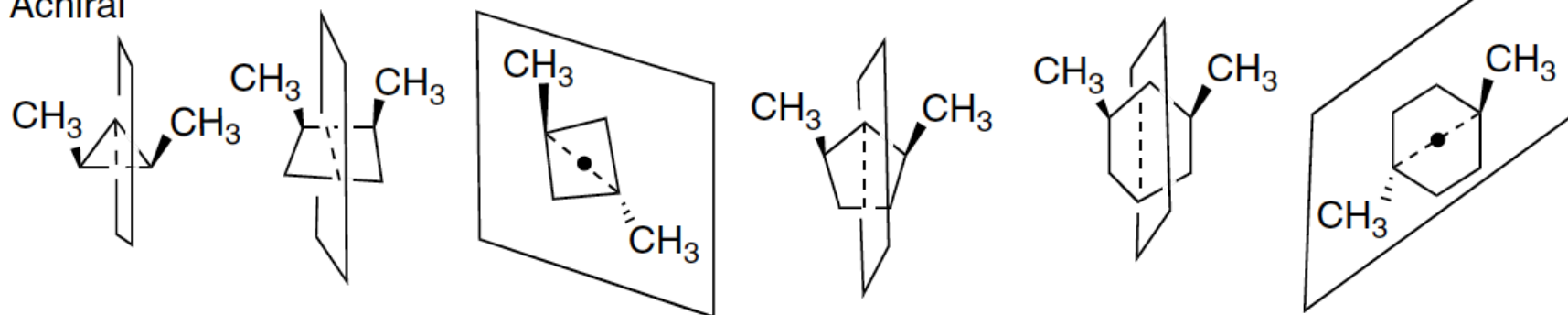


II

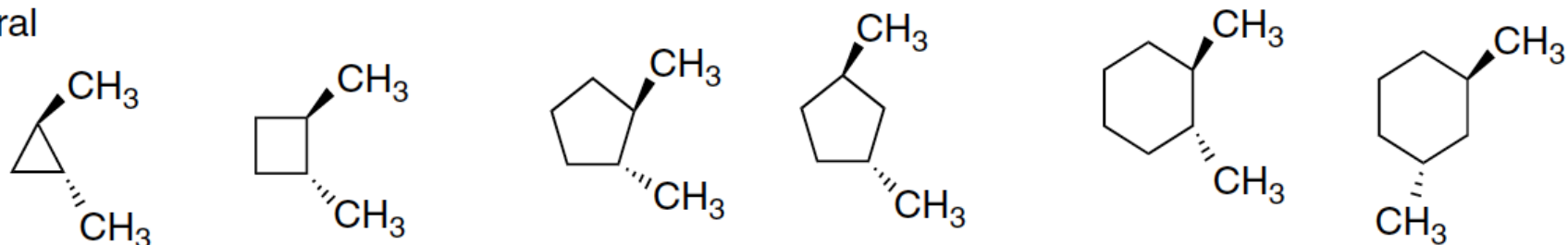
Ra  
Sa

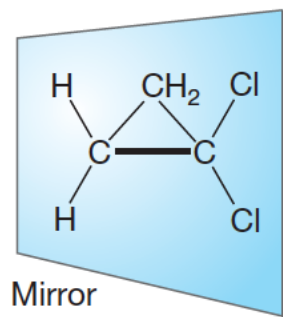
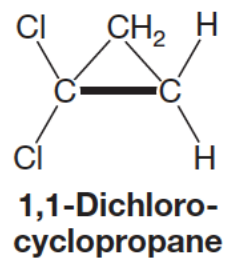
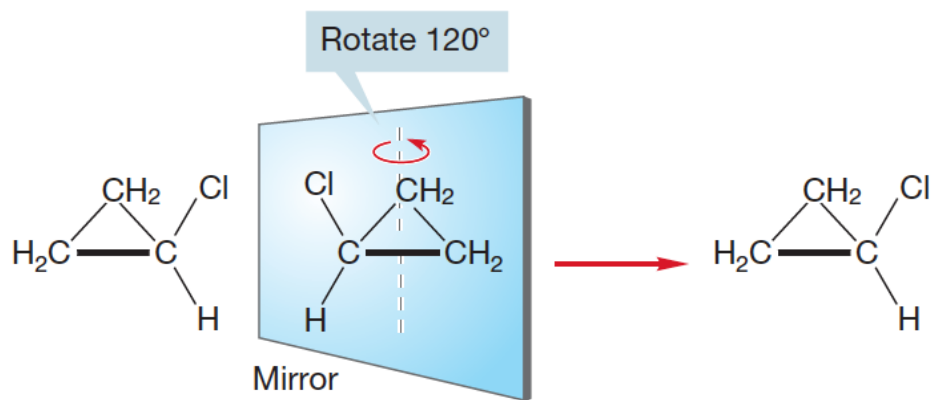
## Scheme 2.1. Chiral and Achiral Disubstituted Cycloalkanes

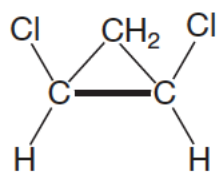
Achiral



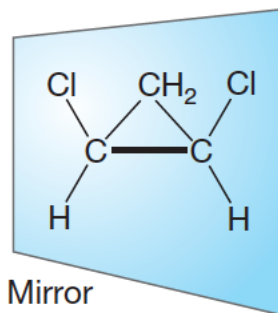
Chiral



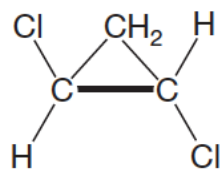




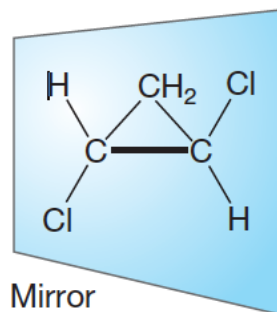
***cis*-1,2-Dichloro-  
cyclopropane**  
(a meso compound)



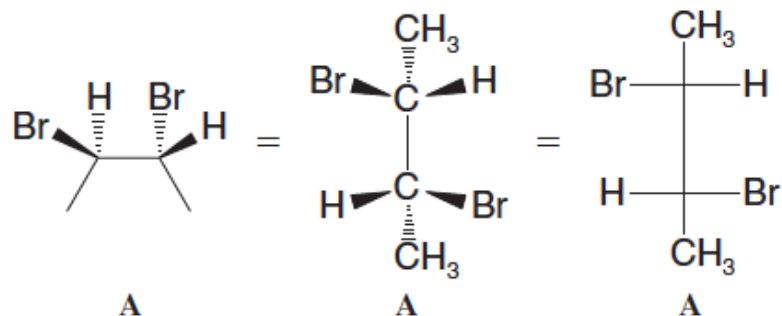
Mirror



***trans*-1,2-Dichloro-  
cyclopropane**



Mirror



Fischer  
projection  
formula

